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IMBEDDING OF JORDAN ALGEBRAS INTO LIE ALGEBRAS. I.

By MAX KOECHER.*

Introduction. Let $\mathfrak{A}$ be a Jordan algebra over a given ground field of characteristic different from two and suppose that $\mathfrak{A}$ contains an unit element. We consider a certain natural construction that associates a Lie algebra $\mathfrak{L}(\mathfrak{A})$ with $\mathfrak{A}$. The algebra $\mathfrak{A}$ is "imbedded" in $\mathfrak{L}(\mathfrak{A})$ in the sense that $\mathfrak{A}$ is a subspace of $\mathfrak{L}(\mathfrak{A})$ such that the product of $\mathfrak{A}$ can be expressed in terms of the product in the Lie algebra $\mathfrak{L}(\mathfrak{A})$. As a first property we prove that $\mathfrak{L}$ carries simple algebras into simple algebras. Then by checking the dimensions we shall see for instance that $\mathfrak{L}(\mathfrak{A})$ is a Lie algebra of type E_7 whenever $\mathfrak{A}$ is a central simple exceptional Jordan algebra.

In the case of finite dimensional Jordan algebra $\mathfrak{A}$ we express the radical of $\mathfrak{L}(\mathfrak{A})$ in terms of the radical of $\mathfrak{A}$ and we give some applications of it to the Wedderburn principal theorem for Jordan algebras and the complete reducibility of Jordan modules. Finally associative bilinear forms are constructed.

In the second part of this paper we shall consider another construction of Lie algebras using an automorphism J of the Jordan algebra $\mathfrak{A}$ satisfying $J^2 = Id$. We will show in a forthcoming paper that over the field of real numbers one covers by this second contruction all Lie algebras that belong to bounded symmetric domains in the sense of E. Cartan. There is another way to relate Lie algebras and Jordan algebras which is given by J. L. Kantor [4].

1. A construction of anti-commutative algebras.

1. In the sequel the algebras which we consider are algebras over a given field of characteristic different from two. We say the algebra $\mathfrak{h}$ *acts on the vector space* $\mathfrak{a}$ if there is a bilinear map $(h, a) \to ha$ of $\mathfrak{h} \times \mathfrak{a}$ into $\mathfrak{a}$. A subspace $\mathfrak{c}$ of $\mathfrak{a}$ is called *invariant under* $\mathfrak{h}$ if $\mathfrak{h}\mathfrak{c} \subset \mathfrak{c}$.

We consider triples $(\mathfrak{h}; \mathfrak{a}, \mathfrak{b})$ satisfying the following conditions:

(i) $\mathfrak{h}$ is an anti-commutative algebra that acts on the vector spaces $\mathfrak{a}$ and $\mathfrak{b}$,

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(ii) there exists a pairing of $\mathfrak{a} \times \mathfrak{b}$ in $\mathfrak{h}$, i.e. a bilinear map $(a, b) \to a \square b$ of $\mathfrak{a} \times \mathfrak{b}$ into $\mathfrak{h}$.

The product in the algebra $\mathfrak{h}$ is denoted by $[h_1, h_2]$. Forming the direct sum $\mathfrak{L} = \mathfrak{h} \oplus \mathfrak{a} \oplus \mathfrak{b}$ of the vector spaces we write in an obvious manner the elements of $\mathfrak{L}$ in the form $x = h \oplus a \oplus b$.

Next we define a product $[x_1, x_2]$ in $\mathfrak{L}$ as follows:

$$[x_1, x_2] = [h_1 \oplus a_1 \oplus b_1, h_2 \oplus a_2 \oplus b_2] = h \oplus a \oplus b,$$

where

$$(1.1) \qquad h = [h_1, h_2] + a_1 \square b_2 - a_2 \square b_1, \quad a = h_1 a_2 - h_2 a_1, \quad b = h_1 b_2 - h_2 b_1.$$

Note that the product $[h_1, h_2]$ in $\mathfrak{L}$ coincides with the product in $\mathfrak{h}$ and that the product of h and a resp. b coincides in $\mathfrak{L}$ with the action of h on a resp. b. Moreover one has $[a, b] = -[b, a] = a \square b$. Obviously, $\mathfrak{L}$ is an anti-commutative algebra having $\mathfrak{h}$ as subalgebra.

As a consequence of (1.1) we observe the following composition rules:

$$(1.2) \qquad [\mathfrak{h}, \mathfrak{h}] \subset \mathfrak{h}, [\mathfrak{h}, \mathfrak{a}] \subset \mathfrak{a}, [\mathfrak{h}, \mathfrak{b}] \subset \mathfrak{b}, \quad [\mathfrak{a}, \mathfrak{a}] = [\mathfrak{b}, \mathfrak{b}] = 0, [\mathfrak{a}, \mathfrak{b}] = \mathfrak{a} \square \mathfrak{b} \subset \mathfrak{h}.$$

Here, for instance, $[\mathfrak{h}, \mathfrak{a}]$ stands for the vector space spanned by the elements $[h, a]$, $h \in \mathfrak{h}$, $a \in \mathfrak{a}$.

Changing the pairing $a \square b$ into $\chi(a \square b)$, where χ is a non zero element in the ground field, does not change the type of the algebra $\mathfrak{L}$ since $h \oplus a \oplus b \to h \oplus a \oplus \chi b$ defines an isomorphism of the changed one onto the first one.

Putting $\mathfrak{m} = \mathfrak{a} \oplus \mathfrak{b}$, there is an obvious action of $\mathfrak{h}$ on $\mathfrak{m}$ and we obtain $\mathfrak{L} = \mathfrak{h} \oplus \mathfrak{m}$, $[\mathfrak{h}, \mathfrak{m}] \subset \mathfrak{m}$, $[\mathfrak{m}, \mathfrak{m}] \subset \mathfrak{h}$.

2. Define the projections

$$H : \mathfrak{L} \to \mathfrak{h}, \quad A : \mathfrak{L} \to \mathfrak{a}, \quad B : \mathfrak{L} \to \mathfrak{b},$$

of the vector space $\mathfrak{L}$ by

$$Hx = h, \qquad Ax = a, \qquad Bx = b,$$

whenever $x = h \oplus a \oplus b$. From (1.1) it follows that

$$(1.3) \qquad H[h, x] = [h, Hx], \quad A[h, x] = [h, Ax], \quad B[h, x] = [h, Bx]$$

for all $x \in \mathfrak{L}$ and $h \in \mathfrak{h}$.

Let $\mathfrak{M}$ be an ideal of $\mathfrak{L}$. Then the formulas (1.3) imply that $H\mathfrak{M}$ *is an ideal in* $\mathfrak{h}$ *and that* $A\mathfrak{M}$ *and* $B\mathfrak{M}$ *are invariant under* $\mathfrak{h}$. From (1.1) follows

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$$[x_1, a \oplus b] = (a_1 \square b - a \square b_1) \oplus h_1 a \oplus h_1 b \in \mathfrak{M}$$

for $x_1 = h_1 \oplus a_1 \oplus b_1 \in \mathfrak{M}$ and all $a \in \mathfrak{a}$ and $b \in \mathfrak{b}$. Therefore we obtain

$$(1.4) \qquad (H\mathfrak{M})\mathfrak{b} \subset A\mathfrak{M}, \qquad (H\mathfrak{M})\mathfrak{b} \subset B\mathfrak{M},$$

$$(1.5) \qquad (A\mathfrak{M}) \square \mathfrak{b} \subset H\mathfrak{M}, \qquad \mathfrak{a} \square (B\mathfrak{M}) \subset H\mathfrak{M}.$$

Conversely consider an ideal $\mathfrak{h}_0$ in $\mathfrak{h}$ and $\mathfrak{h}$-invariant subspaces $\mathfrak{a}_0$ and $\mathfrak{b}_0$ of $\mathfrak{a}$ resp. $\mathfrak{b}$ satisfying

$$(1.6) \qquad \mathfrak{h}_0\mathfrak{a} \subset \mathfrak{a}_0, \ \mathfrak{h}_0\mathfrak{b} \subset \mathfrak{b}_0, \mathfrak{a}_0 \square \mathfrak{b} \subset \mathfrak{h}_0, \ \mathfrak{a} \square \mathfrak{b}_0 \subset \mathfrak{h}_0.$$

Then $\mathfrak{h}_0 \oplus \mathfrak{a}_0 \oplus \mathfrak{b}_0$ *is an ideal of* $\mathfrak{L}$.

Hence for any ideal $\mathfrak{M}$ in $\mathfrak{L}$ we may define another ideal by

$$\hat{\mathfrak{M}} = H\mathfrak{M} \oplus A\mathfrak{M} \oplus B\mathfrak{M}.$$

Obviously $\mathfrak{M} \subset \hat{\mathfrak{M}}$. However, in general we have $\mathfrak{M} \neq \hat{\mathfrak{M}}$. An ideal $\mathfrak{M}$ is said to be *split* if $\mathfrak{M} = H\mathfrak{M} \oplus A\mathfrak{M} \oplus B\mathfrak{M}$, i.e. if $\mathfrak{M} = \hat{\mathfrak{M}}$.

Using that product in $\mathfrak{L}$ we define for a given $v \in \mathfrak{L}$ a linear map U_v of $\mathfrak{L}$ into itself by

$$2U_v x = [[v, x], v].$$

For $a \in \mathfrak{a}$ resp. $b \in \mathfrak{b}$ one verifies

$$(1.7) \qquad 2U_a x = (a \square Bx)a \text{ resp. } 2U_b x = -(Ax \square b)b.$$

In particular, $U_a : \mathfrak{b} \to \mathfrak{a}$ and $U_b : \mathfrak{a} \to \mathfrak{b}$.

3. Next we consider some further conditions for the triple $(\mathfrak{h}; \mathfrak{a}, \mathfrak{b})$:

(iii) there exists an element $h_0 \in \mathfrak{h}$ such that $h_0 a = a$, $h_0 b = b$ and $[h_0, h] = 0$ for all $a \in \mathfrak{a}$, $b \in \mathfrak{b}$ and $h \in \mathfrak{h}$,

(iv) the pairing $\mathfrak{a} \square \mathfrak{b} \subset \mathfrak{h}$ admits only trivial annihilators, i.e. $\mathfrak{a} \square \mathfrak{b} = 0$ for $b \in \mathfrak{b}$ implies $a = 0$ and $a \square \mathfrak{b} = 0$ for $a \in \mathfrak{a}$ implies $b = 0$.

(v) $h\mathfrak{a} = 0$ or $h\mathfrak{b} = 0$ for $h \in \mathfrak{h}$ implies $h = 0$.

(vi) the algebra $\mathfrak{h}$ is generated by the subspace $\mathfrak{a} \square \mathfrak{b}$.

From (v) follows that $\mathfrak{h}$ can be considered as a subspace of $\mathrm{Hom}(\mathfrak{a}, \mathfrak{a})$ and of $\mathrm{Hom}(\mathfrak{b}, \mathfrak{b})$. But the product in $\mathfrak{h}$ does not necessarily coincide with the Lie bracket of the linear transformations.

LEMMA 1. *Suppose that* (i)-(iii) *are satisfied by the triple* $(\mathfrak{h}; \mathfrak{a}, \mathfrak{b})$. *Then any ideal of the algebra* $\mathfrak{L} = \mathfrak{h} \oplus \mathfrak{a} \oplus \mathfrak{b}$ *splits.*

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Proof. Choosing h_0 as indicated in (iii) we calculate $[h_0, x] = a \oplus -b$, whenever $x = h \oplus a \oplus b$. Therefore if x belongs to an ideal $\mathfrak{M}$ of $\mathfrak{L}$, $a \oplus -b$ and consequently $x + (a \oplus -b) = h \oplus 2a$ belongs to $\mathfrak{M}$. Again $[h_0, h \oplus 2a] = 2a$ leads to $a \in \mathfrak{M}$. Since a similar argument yields $b \in \mathfrak{M}$ we get also $h \in \mathfrak{M}$. Hence $x = h \oplus a \oplus b \in \mathfrak{M}$ implies $h \in \mathfrak{M}, a \in \mathfrak{M}, b \in \mathfrak{M}$ and consequently $\hat{\mathfrak{M}} = H\mathfrak{M} \oplus A\mathfrak{M} \oplus B\mathfrak{M} \subset \mathfrak{M}$.

Assuming (iv) and using (1.5) we see that $H\mathfrak{M} = 0$ implies $A\mathfrak{M} = B\mathfrak{M} = 0$. *Hence for any ideal $\mathfrak{M}$ of $\mathfrak{L}$ the condition $H\mathfrak{M} = 0$ is equivalent to $\mathfrak{M} = 0$.* Similarly assuming (v) and using (1.4) we see *that $A\mathfrak{M} = 0$ resp. $B\mathfrak{M} = 0$ is equivalent to $\mathfrak{M} = 0$.*

As a consequence of (iii) we obtain

$$\mathfrak{h}\mathfrak{a} = \mathfrak{a}, \quad \mathfrak{h}\mathfrak{b} = \mathfrak{b}. \tag{1.8}$$

LEMMA 2. *Suppose that* (i)-(iv) *are satisfied by the triple* $(\mathfrak{h}; \mathfrak{a}, \mathfrak{b})$ *and let* $\mathfrak{h}$ *be a simple algebra. Then* $\mathfrak{L}$ *is simple.*

Proof. We start with an ideal $\mathfrak{M} \neq 0$ in $\mathfrak{L}$. Hence $H\mathfrak{M} \neq 0$ and since $H\mathfrak{M}$ is an ideal in $\mathfrak{h}$ we end up with $H\mathfrak{M} = \mathfrak{h}$. From (1.4) and (1.8) we conclude $A\mathfrak{M} = \mathfrak{a}$ and $B\mathfrak{M} = \mathfrak{b}$, therefore $\hat{\mathfrak{M}} = \mathfrak{L}$. But Lemma 1 yields that $\mathfrak{M}$ splits, i.e. that $\mathfrak{M} = \hat{\mathfrak{M}}$ holds.

The vector space $\mathfrak{a}$ is called $\mathfrak{h}$-irreducible, if 0 and $\mathfrak{a}$ are the only subspaces of $\mathfrak{a}$ which are invariant under $\mathfrak{h}$. Using this notion we prove

LEMMA 3. *Suppose that* (i)-(vi) *are satisfied by the triple* $(\mathfrak{h}; \mathfrak{a}, \mathfrak{b})$ *and let* $\mathfrak{a}$ *and* $\mathfrak{b}$ *be* $\mathfrak{h}$*-irreducible. Then* $\mathfrak{L}$ *is simple.*

Proof. Let $\mathfrak{M}$ be an ideal in $\mathfrak{L}$. Since $A\mathfrak{M}$ and $B\mathfrak{M}$ are invariant subspaces of $\mathfrak{a}$ resp. $\mathfrak{b}$, it follows that $A\mathfrak{M}$ resp. $B\mathfrak{M}$ are 0 or $\mathfrak{a}$ resp. $\mathfrak{b}$.

Consider the case $A\mathfrak{M} = 0$. Using (1.4) we find $(H\mathfrak{M})\mathfrak{a} = 0$, and using (v) we obtain $H\mathfrak{M} = 0$. But we already know that $H\mathfrak{M} = 0$ implies $\mathfrak{M} = 0$. Similarly $B\mathfrak{M} = 0$ implies $\mathfrak{M} = 0$.

In the remaining case $A\mathfrak{M} = \mathfrak{a}, B\mathfrak{M} = \mathfrak{b}$, again (1.5) leads to $\mathfrak{a} \square \mathfrak{b} \subset H\mathfrak{M}$. Since $H\mathfrak{M}$ is a subalgebra of $\mathfrak{h}$, we conclude from (vi) that $\mathfrak{h} \subset H\mathfrak{M}$. Therefore $\hat{\mathfrak{M}} = \mathfrak{L}$ and again Lemma 1 leads to $\mathfrak{M} = \mathfrak{L}$.

4. We consider two algebras $\mathfrak{L} = \mathfrak{h} \oplus \mathfrak{a} \oplus \mathfrak{b}$ and $\mathfrak{L}' = \mathfrak{h}' \oplus \mathfrak{a}' \oplus \mathfrak{b}'$ for for which there are given three linear mappings

$$\eta: \mathfrak{h} \to \mathfrak{h}', \quad \alpha: \mathfrak{a} \to \mathfrak{a}', \quad \beta: \mathfrak{b} \to \mathfrak{b}'.$$

The triple $(\eta: \alpha, \beta)$ induces a linear map $\phi: \mathfrak{L} \to \mathfrak{L}'$ by

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$$\phi(h \oplus a \oplus b) = h^\eta \oplus a^\alpha \oplus b^\beta,$$

which we denote by $\phi = \eta \oplus \alpha \oplus \beta$. From (1.1) follows immediately, *that* ϕ *is a homomorphism of* $\mathfrak{L}$ *into* $\mathfrak{L}'$ *if and only if* $\eta: \mathfrak{h} \to \mathfrak{h}'$ *is a homomorphism and if, in addition,*

$$(1.9) \qquad (a \square b)^\eta = a^\alpha \square b^\beta, \quad (ha)^\alpha = h^\eta a^\alpha, \quad (hb)^\beta = h^\eta b^\beta,$$

are satisfied.

5. In order to check under which circumstances the algebra

$$\mathfrak{L} = \mathfrak{h} \oplus \mathfrak{a} \oplus \mathfrak{b}$$

is a Lie algebra we consider the Jacobian

$$J(x_1, x_2, x_3) = [[x_1, x_2], x_3] + [[x_2, x_3], x_1] + [[x_3, x_1], x_2]$$

for $x_1, x_2, x_3 \in \mathfrak{L}$. Note that J vanishes if two arguments coincide and that consequenty J alternates in all three variables.

From (1.2) and, in particular, from $[\mathfrak{a}, \mathfrak{a}] = [\mathfrak{b}, \mathfrak{b}] = 0$ follows

$$J(\mathfrak{h}, \mathfrak{a}, \mathfrak{a}) = J(\mathfrak{h}, \mathfrak{b}, \mathfrak{b}) = J(\mathfrak{a}, \mathfrak{a}, \mathfrak{a}) = J(\mathfrak{b}, \mathfrak{b}, \mathfrak{b}) = 0.$$

Since $\mathfrak{h}$ is a subalgebra of $\mathfrak{L}$ we may assume that $\mathfrak{h}$ is a Lie algebra. Using $J(h_1, h_2, a) = [h_1, h_2]a - h_1(h_2 a) + h_2(h_1 a)$ we see that $J(\mathfrak{h}, \mathfrak{h}, \mathfrak{a}) = 0$ is equivalent to the fact that $\mathfrak{a}$ is an $\mathfrak{h}$-module. Assuming that $\mathfrak{a}$ and $\mathfrak{b}$ are $\mathfrak{h}$-modules we see that $\mathfrak{L}$ is a Lie algebra if and only if $J(\mathfrak{h}, \mathfrak{a}, \mathfrak{b}) = J(\mathfrak{a}, \mathfrak{a}, \mathfrak{b}) = J(\mathfrak{a}, \mathfrak{b}, \mathfrak{b}) = 0$. Rewriting these conditions we obtain by using (1.1)

LEMMA 4. *Let* $\mathfrak{h}$ *be a Lie algebra and let* $\mathfrak{a}$ *and* $\mathfrak{b}$ *be* $\mathfrak{h}$*-modules. Then* $\mathfrak{L} = \mathfrak{h} \oplus \mathfrak{a} \oplus \mathfrak{b}$ *is a Lie algebra if and only if both*

$$(1.10) \qquad [h, a \square b] = (ha) \square b + a \square (hb)$$

and

$$(1.11) \qquad (a_2 \square b)a_1 = (a_1 \square b)a_2, \quad (a \square b_1)b_2 = (a \square b_2)b_1$$

for $h \in \mathfrak{h}$ *and the a's in* $\mathfrak{a}$ *and the b's in* $\mathfrak{b}$.

If (1.10) is satisfied by two elements h_1, h_2 of the Lie algebra $\mathfrak{h}$ then one verifies that it is satisfied by $[h_1, h_2]$. Hence we have the

Remark. Under the assumptions of the Lemma the condition (1.10) is satisfied for $h \in \mathfrak{h}$ if it is satisfied for a set of generators of $\mathfrak{h}$.

Let $f \in \mathfrak{a}$ resp. $g \in \mathfrak{b}$ be given elements. We define a product in each vector space $\mathfrak{a}$ and $\mathfrak{b}$ by

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$$a_1 \underset{g}{\perp} a_2 = \tfrac{1}{2}(a_1 \square g)a_2 \text{ resp. } b_1 \underset{f}{\perp} b_2 = \tfrac{1}{2}(f \square b_1)b_2,$$

where $a_i \in \mathfrak{a}$ and $b_i \in \mathfrak{b}$. These products are commutative if (1.11) is satisfied. Moreover (1.7) leads to

$$a \underset{g}{\perp} a = U_a g \text{ resp. } b \underset{f}{\perp} b = U_b f. \tag{1.12}$$

2. The algebras $\mathfrak{L}(\mathfrak{A})$.

1. Let $\mathfrak{A}$ be a commutative algebra. Denote by $L(a)$ the left multiplication by $a \in \mathfrak{A}$, i.e. put $L(a)x = ax$ for $a, x \in \mathfrak{A}$, and put

$$L(\mathfrak{A}) = \{L(a)\,; a \in \mathfrak{A}\}.$$

Let $\mathfrak{h}(\mathfrak{A})$ be the Lie algebra of linear transformations which is generated by $L(a)$ for $a \in \mathfrak{A}$, i.e.

$$\mathfrak{h}(\mathfrak{A}) = L(\mathfrak{A}) + [L(\mathfrak{A}), L(\mathfrak{A})] + \cdots.$$

If $\mathfrak{A}$ has an unit element e, then the sum of the vector spaces $L(\mathfrak{A})$ and $[L(\mathfrak{A}), L(\mathfrak{A})]$ is direct.

The algebra $\mathfrak{A}$ is called *admissible* if the identity map of $L(\mathfrak{A})$ can be extended to an antiautomorphism $T \to T^*$ of the Lie algebra $\mathfrak{h}(\mathfrak{A})$. If $h^-(\mathfrak{A})$ resp. $h^+(\mathfrak{A})$ denote the subspaces of $\mathfrak{h}(\mathfrak{A})$ spanned by the Lie products of an odd resp. even number of transformations $L(a)$, then $T^* = T$ on $h^-(\mathfrak{A})$ and $T^* = -T$ on $h^+(\mathfrak{A})$. Hence $\mathfrak{h}^+(\mathfrak{A}) \cap \mathfrak{h}^-(\mathfrak{A}) = 0$. Conversely, if $\mathfrak{h}^+(\mathfrak{A}) \cap \mathfrak{h}^-(\mathfrak{A}) = 0$ then $\mathfrak{A}$ is admissible.

There are at least two types of admissible algebras: First the commutative algebras $\mathfrak{A}$ with unit element for which $L(\mathfrak{A})$ is a Lie triple system and second the finite dimensional commutative algebras $\mathfrak{A}$ for which there exists a non degenerate bilinear form σ such that $\sigma(ab, c) = \sigma(a, bc)$ for $a, b, c \in \mathfrak{A}$. Indeed, in the first case we have $\mathfrak{h}^-(\mathfrak{A}) = L(\mathfrak{A})$, $\mathfrak{h}^+(\mathfrak{A}) = [L(\mathfrak{A}), L(\mathfrak{A})]$, and in the second case for $T \in \mathfrak{h}(\mathfrak{A})$ the adjoint transformation T^* (with respect to σ) is well defined and extends the identity map of $L(\mathfrak{A})$ to $\mathfrak{h}(\mathfrak{A})$. But there are power associative algebras with unit element which are not admissible.

In order to construct a triple $\mathfrak{L} = (\mathfrak{h}\,; \mathfrak{a}, \mathfrak{b})$ we start with an admissible algebra $\mathfrak{A}$ and put $\mathfrak{h} = \mathfrak{h}(\mathfrak{A})$. Using another copy $\bar{\mathfrak{A}}$ of $\mathfrak{A}$ and an isomorphism $a \to \bar{a}$ of $\mathfrak{A}$ onto $\bar{\mathfrak{A}}$ we put $\mathfrak{a} = \mathfrak{A}$ and $\mathfrak{b} = \bar{\mathfrak{A}}$. For $T \in \mathfrak{h}(\mathfrak{A})$ the action of T on $a \in \mathfrak{A}$ is simply given by Ta, i.e. the evaluation of T on a, but the action of T on $\bar{b} \in \bar{\mathfrak{A}}$ is given by $-\overline{T^* b}$. *With these actions,* $\mathfrak{a}$ *and* $\mathfrak{b}$ *are* $\mathfrak{h}(\mathfrak{A})$*-modules.*

Next we define the pairing $\mathfrak{A} \times \mathfrak{A}$ in $\mathfrak{h}(\mathfrak{A})$ by

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$$(2.1) \qquad a \square b = 2(L(ab) + [L(a), L(b)]), \text{ for } a, b \in \mathfrak{A}.$$

Then a pairing of $\mathfrak{A} \times \bar{\mathfrak{A}}$ in $\mathfrak{h}(\mathfrak{A})$ is simply given by $a \square \bar{b} = a \square b$ for $a, b \in \mathfrak{A}$. Obviously, the conditions (i) and (ii) are satisfied by the triple $(\mathfrak{h}(\mathfrak{A}); \mathfrak{A}, \bar{\mathfrak{A}})$. The anticommutative algebra given by the triple $(\mathfrak{h}(\mathfrak{A}); \mathfrak{A}, \bar{\mathfrak{A}})$ we denote by

$$\mathfrak{L}(\mathfrak{A}) = \mathfrak{h}(\mathfrak{A}) \oplus \mathfrak{A} \oplus \bar{\mathfrak{A}}.$$

Note that $\mathfrak{L}(\mathfrak{A})$ can be defined for arbitrary commutative algebras $\mathfrak{A}$ using a map $T \to T^*$ of $\mathfrak{h}(\mathfrak{A})$ into itself. But in general $\bar{\mathfrak{A}}$ will not be an $\mathfrak{h}(\mathfrak{A})$-module.

From (2.1) we obtain

$$(2.2) \qquad (a \square b)c = (c \square b)a, \quad (a \square b)^* = b \square a, \text{ for } a, b, c \in \mathfrak{A}.$$

Using the map $a \to \bar{a}$ of $\mathfrak{A}$ onto $\bar{\mathfrak{A}}$ we define a linear map $x \to \bar{x}$ of $\mathfrak{L}(\mathfrak{A})$ onto itself by

$$\bar{x} = (-T^*) \oplus b \oplus \bar{a}, \text{ whenever } x = T \oplus a \oplus \bar{b}.$$

One verifies *that* $x \to \bar{x}$ *is an automorphism of the anticommutative algebra* $\mathfrak{L}(\mathfrak{A})$ *satisfying* $\bar{\bar{x}} = x$. The map $x \to \bar{x}$ is called the *main involution of* $\mathfrak{L}(\mathfrak{A})$. We identify the vector spaces $\mathfrak{h}(\mathfrak{A})$, $\mathfrak{A}$ and $\bar{\mathfrak{A}}$ with the corresponding subspaces of $\mathfrak{L}(\mathfrak{A})$.

2. Rewriting the definitions (1.1) we obtain

$$[T_1 \oplus a_1 \oplus \bar{b}_1, T_2 \oplus a_2 \oplus \bar{b}_2] = T \oplus a \oplus \bar{b}$$

where

$$(2.3) \quad T = [T_1, T_2] + a_1 \square b_2 - a_2 \square b_1, \; a = T_1 a_2 - T_2 a_1, \; b = T_2^* b_1 - T_1^* b_2,$$

and $T_i \in \mathfrak{h}(\mathfrak{A})$, $a_i, b_i \in \mathfrak{A}$.

LEMMA 5. *Let* $\mathfrak{A}$ *be a commutative admissible algebra with unit element* e. *Then the conditions* (i)-(vi) *are satisfied by* $\mathfrak{L}(\mathfrak{A})$.

Proof. From the definition of $\mathfrak{h}(\mathfrak{A})$ follows that $L(e) = Id$ belongs to $\mathfrak{h}(\mathfrak{A})$. Thus (iii) is satisfied for $h_0 = Id$.

Since $a \square b = 0$ implies $e \square b = 2L(b) = 0$ and therefore $b = 0$, and since a similar argument works for $\mathfrak{b}$, the condition (iv) holds.

The condition (v) is a trivial consequence of the fact that the elements of $\mathfrak{h}(\mathfrak{A})$ are linear transformations of $\mathfrak{A}$.

Finally (vi) follows from $a \square e = 2L(a)$, because $\mathfrak{h}(\mathfrak{A})$ is generated by $L(a)$, $a \in \mathfrak{A}$.

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Forming the product in $\mathfrak{a} = \mathfrak{A}$ as described in § 1.5 with respect to the element $\bar{g}$ we obtain for this product

$$a_1 \underset{\bar{g}}{\perp} a_2 = \tfrac{1}{2}(a_1 \square g)a_2 = a_1(a_2 g) + a_2(a_1 g) - (a_1 a_2)g$$

and a similar expression in $\mathfrak{b} = \bar{\mathfrak{A}}$. Since both products are commutative, the condition (1.11) of Lemma 4 is satisfied for every algebra $\mathfrak{L}(\mathfrak{A})$. Moreover (1.7) yields

$$U_a \bar{g} = P(a)g, \qquad a, g \in \mathfrak{A}, \tag{2.4}$$

where $P(a) = 2L^2(a) - L(a^2)$ denotes the so-called quadratic representation of the algebra $\mathfrak{A}$.

3. A subspace $\mathfrak{m}$ of $\mathfrak{a} = \mathfrak{A}$ is invariant under $\mathfrak{h}(\mathfrak{A})$ if $Tm \in \mathfrak{m}$ for $T \in \mathfrak{h}(\mathfrak{A})$ and $m \in \mathfrak{m}$. Since $T = L(a)$, $a \in \mathfrak{A}$, generate $\mathfrak{h}(\mathfrak{A})$, $\mathfrak{m}$ is invariant if and only if $\mathfrak{m}$ is an ideal in $\mathfrak{A}$. The same argument works for $\mathfrak{b} = \bar{\mathfrak{A}}$.

Let $\mathfrak{m}$ be an ideal of $\mathfrak{A}$ and denote by $\mathfrak{i}(\mathfrak{m})$ the ideal in $\mathfrak{h}(\mathfrak{A})$ that is generated by $L(\mathfrak{m}) = \{L(m), m \in \mathfrak{m}\}$, i.e.

$$\mathfrak{i}(\mathfrak{m}) = L(\mathfrak{m}) + [L(\mathfrak{m}), L(\mathfrak{A})] + \cdots.$$

Using (2.1) we observe that $\mathfrak{m} \square \mathfrak{A} \subset \mathfrak{i}(\mathfrak{m})$ and $\mathfrak{A} \square \mathfrak{m} \subset \mathfrak{i}(\mathfrak{m})$.

On the other hand consider the subspace

$$\mathfrak{j}(\mathfrak{m}) = \{T; T \in \mathfrak{h}(\mathfrak{A}), T\mathfrak{A} \subset \mathfrak{m} \text{ and } T^*\mathfrak{A}\}$$

of $\mathfrak{h}(\mathfrak{A})$. Because of the invariance of $\mathfrak{m}$ under $\mathfrak{h}(\mathfrak{A})$ we see that $\mathfrak{j}(\mathfrak{m})$ is also an ideal in $\mathfrak{h}(\mathfrak{A})$. From $L(\mathfrak{m}) \subset \mathfrak{j}(\mathfrak{m})$ follows $\mathfrak{i}(\mathfrak{m}) \subset \mathfrak{j}(\mathfrak{m})$.

Now setting

$$\tilde{\mathfrak{J}}(\mathfrak{m}) = \mathfrak{i}(\mathfrak{m}) \oplus \mathfrak{m} \oplus \bar{\mathfrak{m}} \text{ and } \mathfrak{J}(\mathfrak{m}) = \mathfrak{j}(\mathfrak{m}) \oplus \mathfrak{m} \oplus \bar{\mathfrak{m}}$$

we obtain two subspaces of $\mathfrak{L}(\mathfrak{A})$. But because of (1.6) *both are ideals in* $\mathfrak{L}(\mathfrak{A})$. Obviously, $\tilde{\mathfrak{J}}(\mathfrak{m})$ coincides with the ideal generated by $L(\mathfrak{m})$, while $\mathfrak{J}(\mathfrak{m})$ is the biggest ideal $\mathfrak{M}$ in $\mathfrak{L}(\mathfrak{A})$ such that $A\mathfrak{M} = B\mathfrak{M} = \mathfrak{m}$.

Since $\mathfrak{m} \neq 0$, $\mathfrak{A}$ implies $\tilde{\mathfrak{J}}(\mathfrak{m}) \neq 0$, $\mathfrak{L}(\mathfrak{A})$, we see that $\mathfrak{A}$ is simple if $\mathfrak{L}(\mathfrak{A})$ it is. Conversely since $\mathfrak{a} = \mathfrak{A}$ and $\mathfrak{b} = \bar{\mathfrak{A}}$ both are $\mathfrak{h}(\mathfrak{A})$-irreducible if $\mathfrak{A}$ is simple, a combination of Lemma 3 and Lemma 5 shows that $\mathfrak{L}(\mathfrak{A})$ is simple if $\mathfrak{A}$ it is. Summing up we obtain:

THEOREM 1. *Let $\mathfrak{A}$ be an admissible commutative algebra with unit element. Then $\mathfrak{L}(\mathfrak{A})$ is simple if and only if $\mathfrak{A}$ is simple.*

Now let $\mathfrak{A}$ be a finite direct sum of ideals $\mathfrak{A}_i$,

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$$\mathfrak{A} = \mathfrak{A}_1 \oplus \mathfrak{A}_2 \oplus \cdots \oplus \mathfrak{A}_q, \quad \mathfrak{A}_i \mathfrak{A}_j = 0 \text{ for } i \neq j.$$

From

$$L(\mathfrak{A}) = L^{(1)}(\mathfrak{A}_1) \oplus \cdots \oplus L^{(q)}(\mathfrak{A}_q), \quad L^{(i)}(\mathfrak{A}_i) L^{(j)}(\mathfrak{A}_j) = 0 \text{ for } i \neq j,$$

where $L^{(i)}$ denotes the left multiplication in $\mathfrak{A}_i$, we obtain

$$\mathfrak{h}(\mathfrak{A}) = \mathfrak{h}(\mathfrak{A}_1) \oplus \cdots \oplus \mathfrak{h}(\mathfrak{A}_q), [\mathfrak{h}(\mathfrak{A}_i), \mathfrak{h}(\mathfrak{A}_j)] = 0 \text{ for } i \neq j.$$

Hence the algebra $\mathfrak{L}(\mathfrak{A})$ can be identified with $\mathfrak{L}(\mathfrak{A}_1) \oplus \cdots \oplus \mathfrak{L}(\mathfrak{A}_q)$. In particular we observe the following

COROLLARY. *Let $\mathfrak{A}$ be an admissible commutative algebra with unit element. If $\mathfrak{A}$ is a direct sum of simple ideals, then $\mathfrak{L}(\mathfrak{A})$ is a direct sum of simple ideals.*

4. The relation between $\mathfrak{A}$ and $\mathfrak{L}(\mathfrak{A})$ is functorial in the following sense: Let $W: \mathfrak{A} \to \mathfrak{A}'$ be a homomorphism of the two commutative algebras, i.e. $WL(u) = L'(Wu)W$, where $L'(u)$ denotes the left multiplication of $\mathfrak{A}'$. Hence W induces a linear map $L(u) \to L^W(u) = L'(Wu)$ of $L(\mathfrak{A})$ into $L'(\mathfrak{A}')$. Denote by $\mathrm{map}(\mathfrak{A}, \mathfrak{A}')$ the set of homomorphisms $W: \mathfrak{A} \to \mathfrak{A}'$ for which the map $L(u) \to L^W(u)$ of $L(\mathfrak{A})$ into $L'(\mathfrak{A}')$ can be extended to a homomorphism $T \to T^W$ of the algebra $\mathfrak{h}(\mathfrak{A})$ into $\mathfrak{h}(\mathfrak{A}')$. In this case we get $L^W(u)W = WL(u)$ and

$$[L(u), L(v)]^W \; W = [L^W(u), L^W(v)]W = W[L(u), L(v)]$$

and in general

$$WT = T^W \; W, \qquad T \in \mathfrak{h}(\mathfrak{A}). \tag{2.5}$$

In particular, *every surjective homomorphism $W: \mathfrak{A} \to \mathfrak{A}'$ belongs to* map $(\mathfrak{A}, \mathfrak{A}')$, because of the fact, that any representation of T as a product of some $L(u)$ yields an element $T' \in \mathfrak{h}(\mathfrak{A}')$ such that $WT = T'W$. And W being surjective shows that T' is uniquely determined by T. Hence $T \to T'$ is a homomorphism too. This construction also shows that

$$(T^W)^* = (T^*)^W, \qquad T \in \mathfrak{h}(\mathfrak{A}), \tag{2.6}$$

holds, whenever $\mathfrak{A}$ and $\mathfrak{A}'$ are admissible. Finally we observe for the pairing $\square$ in $\mathfrak{L}(\mathfrak{A})$ resp. $\square'$ in $\mathfrak{L}(\mathfrak{A}')$ that

$$\begin{aligned} \tfrac{1}{2}(a \square b)^W &= L^W(ab) + [L(a), L(b)]^W \\ &= L'(Wa \cdot Wb) + [L'(Wa), L'(Wb)] \\ &= \tfrac{1}{2} Wa \square' Wb. \end{aligned} \tag{2.7}$$

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For any $W \in \mathrm{map}(\mathfrak{A}, \mathfrak{A}')$ we define a map $\mathfrak{L}_W : \mathfrak{L}(\mathfrak{A}) \to \mathfrak{L}(\mathfrak{A}')$ by

$$\mathfrak{L}_W(T \oplus a \oplus \bar{b}) = T^W \oplus Wa \oplus \overline{Wb}.$$

From (2.5)-(2.7) and (1.9) follows *that* $\mathfrak{L}_W$ *is a homomorphism of the anticommutative algebras* $\mathfrak{L}(\mathfrak{A})$ *into* $\mathfrak{L}(\mathfrak{A}')$. Moreover, let $\mathcal{A}$ be the category of admissible commutative algebras, for which the maps are defined as above. Then one verifies that $\mathfrak{L}$ is a covariant functor of the category $\mathcal{A}$ into the category of anticommutative algebras.

Next let $W : \mathfrak{A} \to \mathfrak{A}'$ be a *surjective* homomorphism, hence W is in $\mathrm{map}(\mathfrak{A}, \mathfrak{A}')$. From the construction of T^W for $T \in \mathfrak{h}(\mathfrak{A})$ we see, that $T \to T^W$ is also surjective. *Hence* $\mathfrak{L}_W$ *is surjective too.* Moreover $\mathfrak{L}_W(T \oplus a \oplus \bar{b}) = 0$ is equivalent to $T^W = 0$ and $a, b \in \mathrm{kernel}\, W$. But using (2.5) and (2.6) one verifies that $T^W = 0$ is equivalent to $T\mathfrak{A} \in \mathrm{kernel}\, W$ and $T^*\mathfrak{A} \in \mathrm{kernel}\, W$. Therefore we obtain

(2.8) $\mathrm{kernel}\, \mathfrak{L}_W = \mathfrak{J}(\mathrm{kernel}\, W)$, whenever $W : \mathfrak{A} \to \mathfrak{A}'$ surjective.

On the other hand let $V : \mathfrak{A}'' \to \mathfrak{A}$ be an *injective* homomorphism in $\mathrm{map}(\mathfrak{A}'', \mathfrak{A})$. Using again (2.5) we see that $T \to T^V$ is injective. *Hence* $\mathfrak{L}_V$ *is injective.* From the definition of $\mathfrak{L}_V$ follows that the image of $\mathfrak{L}(\mathfrak{A}'')$ under $\mathfrak{L}_V$ is given by $\mathfrak{S}(\mathfrak{B}) = \mathfrak{s}(\mathfrak{B}) \oplus \mathfrak{B} \oplus \bar{\mathfrak{B}}$ where $\mathfrak{B}$ is the image of V and where $\mathfrak{s}(\mathfrak{B})$ is the subalgebra of $\mathfrak{h}(\mathfrak{A})$ generated by $L(b)$, $b \in \mathfrak{B}$. Therefore we have

(2.9) $\mathrm{image}\, \mathfrak{L}_V = \mathfrak{S}(\mathrm{image}\, V)$, whenever $V : \mathfrak{A}'' \to \mathfrak{A}$ injective.

Since for an ideal $\mathfrak{m}$ of $\mathfrak{A}$ the algebra $\mathfrak{S}(\mathfrak{m})$ is in general properly contained in the ideal $\mathfrak{J}(\mathfrak{m})$, the functor $\mathfrak{L}$ does *not* carry short exact sequences into short exact sequences.

As a consequence of (2.8) we obtain

$$\mathfrak{L}(\mathfrak{A}/\mathfrak{m}) \cong \mathfrak{L}(\mathfrak{A})/\mathfrak{J}(\mathfrak{m}), \tag{2.10}$$

whenever $\mathfrak{m}$ is an ideal of $\mathfrak{A}$ such that the quotient algebra $\mathfrak{A}/\mathfrak{m}$ is admissible too.

5. Let $\mathrm{Aut}\,\mathfrak{A}$ be the group of automorphisms of the algebra $\mathfrak{A}$. Then $\mathrm{Aut}\,\mathfrak{A}$ is contained in $\mathrm{map}(\mathfrak{A}, \mathfrak{A})$ and consequently $\mathfrak{L}_W \in \mathrm{Aut}\,\mathfrak{L}(\mathfrak{A})$ for every $W \in \mathrm{Aut}\,\mathfrak{A}$.

Next we consider derivations of $\mathfrak{A}$ and $\mathfrak{L}(\mathfrak{A})$. For an arbitrary algebra $\mathfrak{B}$ the Lie algebra of the derivations of $\mathfrak{B}$ is denoted by $\mathrm{Der}\,\mathfrak{B}$.

A derivation D of $\mathfrak{A}$ can be extended to an inner derivation of $\mathfrak{h}(\mathfrak{A})$ by setting $T^D = [D, T]$ for $T \in \mathfrak{h}(\mathfrak{A})$. From $L^D(u) = [D, L(u)] = L(Du)$

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follows that $T \to T^D$ map $\mathfrak{h}(\mathfrak{A})$ into itself, because $\mathfrak{h}(\mathfrak{A})$ is generated by $L(\mathfrak{A})$. For $D \in \operatorname{Der}\mathfrak{A}$ we define

$$\mathfrak{L}_D(T \oplus a \oplus \bar{b}) = T^D \oplus Da \oplus \overline{Db}.$$

Thus $\mathfrak{L}_D$ is a linear map of $\mathfrak{L}(\mathfrak{A})$ into itself. One verifies that $(a \square b)^D = Da \square b + a \square Db$ and $DTa = T^D a + TDa$, $T \in \mathfrak{h}(\mathfrak{A})$. Finally from $(L^D(u))^* = [D, L(u)]^* = L^*(Du) = L(Du) = L^D(u)$ we observe that $(T^D)^* = (T^*)^D$ for $T \in \mathfrak{h}(\mathfrak{A})$. Using these formulas an easy verification shows *that for a derivation D of $\mathfrak{A}$ the map $\mathfrak{L}_D$ is a derivation of the algebra $\mathfrak{L}(\mathfrak{A})$.*

LEMMA 6. *Let D be a derivation of $\mathfrak{A}$. Then $\mathfrak{L}_D$ is an inner derivation of $\mathfrak{L}(\mathfrak{A})$ if and only if $D \in \mathfrak{h}(\mathfrak{A})$ and $D^* = -D$.*

Here by an inner derivation of $\mathfrak{L}(\mathfrak{A})$ we mean as usual a map $x \to [d, x]$, $d \in \mathfrak{L}(\mathfrak{A})$ that is a derivation of $\mathfrak{L}(\mathfrak{A})$.

Proof. From $\mathfrak{L}_D x = [d, x]$, $d = T_0 \oplus a_0 \oplus \bar{b}_0 \in \mathfrak{L}(\mathfrak{A})$, follows in particular $Da = T_0 a - Ta_0$, where $x = T \oplus a \oplus \bar{b}$. Hence $a_0 = 0$ and $D = T_0 \in \mathfrak{h}(\mathfrak{A})$. Similarly we get $b_0 = 0$ and $d = D \in \mathfrak{h}(\mathfrak{A})$. Furthermore $-\overline{D^* b} = [d, \bar{b}] = \mathfrak{L}_D \bar{b} = \overline{Db}$ implies $D^* = -D$.

Conversely, let be $D \in \mathfrak{h}(\mathfrak{A})$ and suppose $D^* = -D$. Then the definition of $\mathfrak{L}_D$ yields $\mathfrak{L}_D x = [D, x]$.

6. Let J be an automorphism of the admissible algebra $\mathfrak{A}$ satisfying $J^2 = Id$. Setting

$$T^J = -JT^*J, \qquad T \in \mathfrak{h}(\mathfrak{A}),$$

we obtain *an automorphism* $T \to T^J$ of $\mathfrak{h}(\mathfrak{A})$. Indeed, from

$$L^J(u) = -JL(u)J = -L(Ju)$$

follows $T^J \in \mathfrak{h}(\mathfrak{A})$ and since $T \to T^*$ is an antiautomorphism of $\mathfrak{h}(\mathfrak{A})$, $T \to T^J$ is an automorphism.

We define subspaces $\mathfrak{h}^+$ resp. $\mathfrak{h}^-$ of $\mathfrak{h}(\mathfrak{A})$ by

$$\mathfrak{h}^\pm = \mathfrak{h}^\pm(\mathfrak{A}; J) = \{T; T \in \mathfrak{h}(\mathfrak{A}), T^J = \pm T\}.$$

Note that in case $J = Id$ the spaces $\mathfrak{h}^\pm$ coincides with the spaces $\mathfrak{h}^\pm(\mathfrak{A})$ introduced in Section 1.

Obviously $\mathfrak{h}^+$ is a subalgebra of $\mathfrak{h}(\mathfrak{A})$ and moreover we obtain a decomposition

$$\mathfrak{h}(\mathfrak{A}) = \mathfrak{h}^+ + \mathfrak{h}^-, \text{ where } [\mathfrak{h}^+, \mathfrak{h}^-] \subset \mathfrak{h}^-, [\mathfrak{h}^-, \mathfrak{h}^-] \subset \mathfrak{h}^+.$$

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Next we define subspaces $\mathfrak{L}^+$ resp. $\mathfrak{L}^-$ of $\mathfrak{L}(\mathfrak{A})$ by setting

$$\mathfrak{L}^{\pm} = \mathfrak{L}^{\pm}(\mathfrak{A}; J) = \{T \oplus a \oplus (\overline{\pm Ja}); T \in \mathfrak{h}^{\pm}(\mathfrak{A}; J), a \in \mathfrak{A}\}.$$

Obviously, as a vector space $\mathfrak{L}^{\pm}$ is isomorphic to $h^{\pm} \oplus \mathfrak{A}$. Again we obtain a direct sum decomposition

$$\mathfrak{L}(\mathfrak{A}) = \mathfrak{L}^+ + \mathfrak{L}^- \tag{2.11}$$

with respect to the given automorphism J of $\mathfrak{A}$. Instead of $\mathfrak{L}^{\pm}$ we write also $\mathfrak{L}^{\epsilon}$, where $\epsilon = \pm 1$. Let $x_i = T_i \oplus a_i \oplus \overline{\epsilon_i J a_i}$ be an element on $\mathfrak{L}^{\epsilon_i}$, $\epsilon_i = \pm 1$. Then one verifies $[x_1, x_2] = T \oplus a \oplus \epsilon_1\epsilon_2 \overline{Ja}$, where

$$T = [T_1, T_2] + \epsilon_2 a_1 \square Ja_2 - \epsilon_1 a_2 \square Ja_1, \quad a = T_1 a_2 - T_2 a_1. \tag{2.12}$$

Indeed, the last component in the product is

$$\epsilon_1 T_2^* Ja_1 - \epsilon_2 T_1^* Ja_2 = \epsilon_1\epsilon_2 (T_1 a_2 - T_2 a_1) = \epsilon_1\epsilon_2 Ja$$

because $T_i^* J = -\epsilon_i J T_i$. Using

$$(a \square Jb)^* J = 2(L(a \cdot Jb) - [L(a), L(Jb)])J = J(b \square Ja)$$

it follows from (2.12) that $T^* J = \epsilon_1\epsilon_2 JT$ is satisfied. Hence $T \in \mathfrak{h}^{\epsilon_1\epsilon_2}$ and $[\mathfrak{L}^{\epsilon_1}, \mathfrak{L}^{\epsilon_2}] \subset \mathfrak{L}^{\epsilon_1\epsilon_2}$. Therefore we end up with a subalgebra $\mathfrak{L}^+$ of $\mathfrak{L}(\mathfrak{A})$ and the composition rules

$$[\mathfrak{L}^+, \mathfrak{L}^-] \subset \mathfrak{L}^-, \quad [\mathfrak{L}^-, \mathfrak{L}^-] \subset \mathfrak{L}^+. \tag{2.13}$$

Hence every automorphism J satisfying $J^2 = Id$ (even $J = Id$) induces a "Cartan-decomposition" (2.11), (2.13) of $\mathfrak{L}(\mathfrak{A})$.

From (2.13) follows that the algebra $\mathfrak{L}^+$ acts on the vector space $\mathfrak{L}^-$. *Moreover, $\mathfrak{L}^+$ acts faithfully on $\mathfrak{L}^-$ because*

$$x \in \mathfrak{L}^+, \ [x, \mathfrak{L}^-] = 0 \text{ implies } x = 0.$$

Indeed, let $x = T_1 \oplus a_1 \oplus \overline{Ja_1}$. Then (2.12) yields $T_1 a_2 = T_2 a_1$ for all $a_2 \in \mathfrak{A}$ and all $T_2 \in \mathfrak{h}^-$. Hence $T_1 = 0$ and $T_2 = Id \in \mathfrak{h}^-$ implies $a_1 = 0$.

3. Jordan algebras.

1. A linear transformation D of the algebra $\mathfrak{A}$ is a derivation if $D(uv) = Du \cdot v + u \cdot Dv$ holds for $u, v \in \mathfrak{A}$. This is equivalent to $L(Du) = [D, L(u)]$ for $u \in \mathfrak{A}$. We denote by (u, v, w) the *associator* of the elements $u, v, w \in \mathfrak{A}$, i. e.

$$(u, v, w) = (uv)w - u(vw) = [L(w), L(u)]v.$$

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Now let $\mathfrak{A}$ be an admissible commutative algebra with unit element e. We know that the conditions (1.11) are satisfied for the algebra $\mathfrak{L}(\mathfrak{A})$ (see § 2.2). Rewriting the condition (1.10) and using the remark following Lemma 4 we see that $\mathfrak{L}(\mathfrak{A})$ is a Lie algebra if and only if $[T, a \square b] = [Ta \square b] - [a \square T^* b]$ is satisfied for $T \in L(\mathfrak{A})$ and $a, b \in \mathfrak{A}$, because $L(\mathfrak{A})$ generates $\mathfrak{h}(\mathfrak{A})$. Choosing $T = L(c)$ we obtain

$$[L(c), a \square b] = ac \square b - a \square bc, \qquad a, b, c \in \mathfrak{A}, \tag{3.1}$$

as a necessary and sufficient condition. Adding resp. subtracting the same identity in which a and b are changed, we see that (3.1) is equivalent to

$$[L(a), L(bc)] + [L(b), L(ca)] + [L(c), L(ab)] = 0, \quad a, b, c \in \mathfrak{A}, \tag{A}$$

and

$$[[L(a), L(b)], L(c)] = L([L(a), L(b)]c) = L((b, c, a)), \qquad a, b, c \in \mathfrak{A}. \tag{B}$$

It is well known that (B) is a consequence of (A). *Hence* (3.1) *and* (A) *are equivalent.*

Assuming only (B) for the commutative algebra $\mathfrak{A}$, i.e. assuming that $L(\mathfrak{A})$ is a Lie triple system, we obtain obviously

$$\mathfrak{h}(\mathfrak{A}) = L(\mathfrak{A}) \oplus \mathfrak{d}(\mathfrak{A}), \quad \mathfrak{d}(\mathfrak{A}) = [L(\mathfrak{A}), L(\mathfrak{A})], \tag{3.2}$$

and the sum is a direct sum of the vector spaces. In the notation of § 2.1 we have $\mathfrak{h}^-(\mathfrak{A}) = L(\mathfrak{A})$ and $\mathfrak{h}^+(\mathfrak{A}) = \mathfrak{d}(\mathfrak{A})$. We know already that any such algebra is admissible.

Summing up: $\mathfrak{L}(\mathfrak{A})$ *is a Lie algebra if and only if* (A) *is satisfied for the algebra* $\mathfrak{A}$.

If the characteristic of the ground field is not three, (A) yields $[L(u), L(u^2)] = 0$ and thus $u(u^2 v) = u^2(uv)$. Hence $\mathfrak{A}$ is a Jordan algebra.

Conversely, let $\mathfrak{A}$ be a Jordan algebra. Then by linearizing the basic identity $u(u^2 v) = u^2(uv)$ one observes (A). We have proved

THEOREM 2. *Let* $\mathfrak{A}$ *be a Jordan algebra. Then* $\mathfrak{L}(\mathfrak{A})$ *is a Lie algebra.*

Let $\mathfrak{J}$ be the category of Jordan algebras over the given ground field and for which map$(\mathfrak{A}, \mathfrak{A}')$ is defined as in § 2.3. *Then* $\mathfrak{L}$ *is a functor of* $\mathfrak{J}$ *into the category of Lie algebras.*

According to J. G. M. Mars one gets $\mathfrak{L}(\mathfrak{A})$ from a generalization of a construction due to J. Tits.

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2. For the results on Jordan algebras we next have to use see N. Jacobson [3], Chapter I, or in the finite dimensional case H. Braun and M. Koecher [1], Chapter V. Let $\mathfrak{A}$ be a Jordan algebra with unit element e and let $f \in \mathfrak{A}$. We consider the *mutation* (= *homotope* resp. *isotope*) $\mathfrak{A}_f$ of $\mathfrak{A}$, which is a Jordan algebra defined in the same vector space as $\mathfrak{A}$ but having the product

$$u \perp v = u(vf) + v(uf) - f(uv) = \{u, f, v\}.$$

We denote the left multiplication of $\mathfrak{A}_f$ by $L_f(u)$, i.e. we put

$$L_f(u) = L(uf) + [L(u), L(f)] = \tfrac{1}{2} u \,\square\, f..$$

Hence the Lie algebra $\mathfrak{h}(\mathfrak{A}_f)$ is a subalgebra of $\mathfrak{h}(\mathfrak{A})$.

Since $\mathfrak{A}_f$ has an unit element if and only if f is invertible in $\mathfrak{A}$, we have to restrict ourself to the case that f is an invertible element of $\mathfrak{A}$. Then the inverse of f is the unit element of $\mathfrak{A}_f$. Since $\mathfrak{A}$ is also a mutation of $\mathfrak{A}_f$ we obtain $\mathfrak{h}(\mathfrak{A}) \subset \mathfrak{h}(\mathfrak{A}_f)$ and consequently $\mathfrak{h}(\mathfrak{A}) = \mathfrak{h}(\mathfrak{A}_f)$.

Next we construct the map $\phi = Id \oplus Id \oplus \beta$ (see § 1.4) of $\mathfrak{L}(\mathfrak{A}_f)$ onto $\mathfrak{L}(\mathfrak{A})$ by setting

$$\phi(T \oplus a \oplus \bar{b}) = T \oplus a \oplus \overline{P(f)b}, \quad b^\beta = P(f)b,$$

where $T \in \mathfrak{h}(\mathfrak{A}_f)$, $a, b \in \mathfrak{A}_f$ and where $P(f) = 2L^2(f) - L(f^2)$.

THEOREM 3. *Let $\mathfrak{A}$ be a Jordan algebra with unit element and let f be an invertible element of $\mathfrak{A}$. Then $\phi: \mathfrak{L}(\mathfrak{A}_f) \to \mathfrak{L}(\mathfrak{A})$ is an isomorphism of the algebras.*

Proof. Denote the antiautomorphism of $\mathfrak{h}(\mathfrak{A})$ by * and the corresponding antiautomorphism of $\mathfrak{h}(\mathfrak{A}_f)$ by #. Using (1.9) we have only to prove

$$a \underset{f}{\square} b = a \,\square\, P(f)b, \quad P(f)T^{\#}b = T^{*}P(f)b, \tag{3.3}$$

for $a, b \in \mathfrak{A}_f$ and $T \in \mathfrak{h}(\mathfrak{A}_f)$. Here $\underset{f}{\square}$ stands for the pairing given by $\mathfrak{A}_f$. Calculating

$$\tfrac{1}{2}(a \underset{f}{\square} b)u = (L_f(a \perp b) + [L_f(u), L_f(b)])u = a \perp (u \perp b) + u \perp (a \perp b) - b \perp (a \perp u)$$

$$= P_f(a, u)b = P(a, u)P(f)b = \tfrac{1}{2}(a \,\square\, P(f)b)u,$$

because the quadratic representation $P_f(u)$ of $\mathfrak{A}_f$ is given by $P(u)P(f)$ and where $P(a, u)$ resp. $P_f(a, u)$ stands for the linearized quadratic representation. Thus the first formula in (3.3) holds. To prove the second formula it is

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enough to prove it for $T = L_f(u)$, because these transformations generate $\mathfrak{h}(\mathfrak{A}_f)$. Since $\#$ is the identity on $L_f(u)$ and since $L_f{}^*(u) = L_u(f)$ follows from the definition of $L_f(u)$, it suffices to check

$$P(f)L_f(u) = L_u(f)P(f)$$

for $u, f \in \mathfrak{A}$. But this is a consequence of MacDonald's theorem.

Since $P(f)$ is bijective the map ϕ is bijective too.

3. Let $c \in \mathfrak{A}$ and $z = T \oplus a \oplus \bar{b}$ be an element of $\mathfrak{L}(\mathfrak{A})$. The condition $[z, c] = \lambda c$ is equivalent to $Tc = \lambda c$, $c \square b = 0$, therefore to

$$Tc = \lambda c,\ cb = 0,\ [L(c), L(b)] = 0. \tag{3.4}$$

As a consequence, $[z, \mathfrak{A}] = 0$ is equivalent to $T = 0$, $a = 0$, and therefore to $z \in \bar{\mathfrak{A}}$. Using the main involution of $\mathfrak{L}(\mathfrak{A})$ we observe that $[z, \bar{\mathfrak{A}}] = 0$ implies $z \in \mathfrak{A}$. Hence $[z, \mathfrak{A} \oplus \bar{\mathfrak{A}}] = 0$ *if and only if* $z = 0$. *In particular, the Lie algebra* $\mathfrak{L}(\mathfrak{A})$ *has center* 0.

Using the normalizer $\mathfrak{N}(c) = \{z; [z, c] = \lambda c\}$ of c we see that $z \in \mathfrak{N}(c)$ and (3.4) are equivalent.

LEMMA 7. *If* $c \in \mathfrak{A}$ *is invertible, then* $\mathfrak{N}(c) \subset \mathfrak{h}(\mathfrak{A}) \oplus \mathfrak{A}$. *Conversely, if* $\mathfrak{A}$ *has finite dimension and if* $\mathfrak{N}(c) \subset \mathfrak{h}(\mathfrak{A}) \oplus \mathfrak{A}$, *then* c *is invertible.*

In the case of finite dimensional algebras this Lemma states a characterization of the invertible elements of $\mathfrak{A}$ in terms of the Lie algebra $\mathfrak{L}(\mathfrak{A})$.

Proof. For $z = T \oplus a \oplus \bar{b}$ we use the last condition of (3.4) and apply it to c. Thus $c(cb) = c^2 b$ and the second condition of (3.4) leads to $P(c)b = 0$. Thus c being invertible implies $b = 0$ and $z \in \mathfrak{h}(\mathfrak{A}) \oplus \mathfrak{A}$.

Conversely, if $\mathfrak{A}$ is finite dimensional and c not invertible, then there exists $b \neq 0$ such that $cb = 0$ and $[L(b), L(c)] = 0$.

4. Finally we consider some examples. First let $\mathfrak{A}$ be an associative Jordan algebra. Then $\mathfrak{h}(\mathfrak{A}) = \mathfrak{L}(\mathfrak{A}) = \mathfrak{A}$ and $\mathfrak{L}(\mathfrak{A})$ simply coincides with the split 3-dimensional Lie algebra over $\mathfrak{A}$, considered as an algebra over the ground field of $\mathfrak{A}$.

Nex let $\mathfrak{A}$ be a central simple Jordan algebra over a ground field of characteristic zero and of finite dimension $n > 1$. Let r be the degree of $\mathfrak{A}$, i. e. the degree of the reduced norm of $\mathfrak{A}$ (see [1], Chapter II, § 4.6). Then

$$n = r + \tfrac{1}{2}r(r-1)d,$$

where for the invariant d of $\mathfrak{A}$ in the case $r \geqq 3$ only the integers 1, 2, 4, 8

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(only for $r=3$) occur and $d=n-2$ for $r=2$. Then from the list in [1], page 293, and from [1], Chapter IX, Theorem 5.1, one observes the following dimensions for $\mathfrak{h}(\mathfrak{A})$ according to the cases $d=1,2,4,8$:

$$r^2, \qquad 2r^2-1 \qquad (2r)^2, \qquad 79.$$

Hence from $\dim \mathfrak{L}(\mathfrak{A}) = \dim \mathfrak{h}(\mathfrak{A}) + 2\dim \mathfrak{A}$ we obtain the following dimensions for $\mathfrak{L}(\mathfrak{A})$ in the case $r \geqq 3$:

$$r(2r+1), \qquad (2r-1)(2r+1), \qquad 2r(4r-1), \qquad 133.$$

The case $r=2$ leads to the dimension $\frac{1}{2}(n+1)(n+2)$ for $\mathfrak{L}(\mathfrak{A})$.

In all these cases the Lie algebras $\mathfrak{L}(\mathfrak{A})$ are simple (see Theorem 1). Since there is only one simple Lie algebra of dimension 133 we observe

THEOREM 4. *Let $\mathfrak{A}$ be a central simple exceptional Jordan algebra over a field of characteristic 0. Then $\mathfrak{L}(\mathfrak{A})$ is a simple Lie algebra of type E_7.*

According to the other dimensions one concludes, that $\mathfrak{L}(\mathfrak{A})$ is of type A_{2r-1} if $d=2$ resp. of type D_{2r} if $d=4$. One can prove (See § 5.3) that $\mathfrak{L}(\mathfrak{A})$ is of type C_r if $d=1$ and that in the case $r=2$ one gets the types B_s if $n=2s-1$, $s \geqq 2$, resp. D_s if $n=2s-2$, $s \geqq 3$.

4. The radical of $\mathfrak{L}(\mathfrak{A})$.

1. Let $\mathfrak{B}$ be an arbitrary algebra and $\mathfrak{M}$ be a subspace of $\mathfrak{B}$. As usual we define the subspaces $\mathfrak{M}^k$ and $\mathfrak{M}^{\langle k\rangle}$ of $\mathfrak{B}$ recursively by setting $\mathfrak{M}^0 = \mathfrak{M}^{\langle 0\rangle} = \mathfrak{M}$ and

$$\mathfrak{M}^{k+1} = \mathfrak{M}\mathfrak{M}^k, \qquad \mathfrak{M}^{\langle k+1\rangle} = \mathfrak{M}^{\langle k\rangle}\mathfrak{M}^{\langle k\rangle}.$$

In general $\mathfrak{M}$ being an ideal does not imply that either $\mathfrak{M}^k$ or $\mathfrak{M}^{\langle k\rangle}$ is an ideal. The subspace $\mathfrak{M}$ is called *nilpotent* resp. *solvable*, if there exists an integer k such that $\mathfrak{M}^k = 0$ resp. $\mathfrak{M}^{\langle k\rangle} = 0$.

Now let $\mathfrak{A}$ be a Jordan algebra with unit element e. For any ideal $\mathfrak{m}$ of $\mathfrak{A}$ we consider the ideal $\mathfrak{i}(\mathfrak{m})$ and $\mathfrak{j}(\mathfrak{m})$ of $\mathfrak{h}(\mathfrak{A})$ which are defined in § 2.3. Using the identity (B) and (2.1) we obtain

$$\mathfrak{i}(\mathfrak{m}) = L(\mathfrak{m}) \oplus [L(\mathfrak{m}), L(\mathfrak{A})] = \mathfrak{m} \square \mathfrak{A}. \tag{4.1}$$

Any element T in $\mathfrak{h}(\mathfrak{m})$ can be written as $T = L(a) + D$, where $a \in \mathfrak{A}$ and $D \in \mathfrak{d}(\mathfrak{A}) = [L(\mathfrak{A}), L(\mathfrak{A})]$. Using the unit element e of $\mathfrak{A}$ and $De = 0$ we see that $T\mathfrak{A} \subset \mathfrak{m}$ implies $a \in \mathfrak{m}$. Hence

$$\mathfrak{j}(\mathfrak{m}) = L(\mathfrak{m}) \oplus \mathfrak{d}(\mathfrak{A}) \cap \mathfrak{j}(\mathfrak{m}). \tag{4.2}$$

From (4.1) and (4.2) it follows that both $\mathfrak{i}(\mathfrak{m})$ and $\mathfrak{j}(\mathfrak{m})$ are mapped into themselves under the map $T \to T^*$.

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LEMMA 8. *Let* $\mathfrak{m}$ *be an ideal in* $\mathfrak{A}$. *Then*

a) $[\mathfrak{j}(\mathfrak{m}), \mathfrak{h}(\mathfrak{A})] \subset \mathfrak{i}(\mathfrak{m})$,

b) $(\mathfrak{j}(\mathfrak{m}))^{\langle k+1\rangle} \subset (\mathfrak{i}(\mathfrak{m}))^{\langle k\rangle}$ *for* $k = 1, 2, \cdots$.

Since $\mathfrak{i}(\mathfrak{m})$ is contained in $\mathfrak{j}(\mathfrak{m})$ we also have the

COROLLARY. $\mathfrak{i}(\mathfrak{m})$ *is solvable if and only if* $\mathfrak{j}(\mathfrak{m})$ *is solvable.*

Proof. a) It is enough to show that

$$[D, \mathfrak{h}(\mathfrak{A})] \subset \mathfrak{i}(\mathfrak{m}) \text{ for } D \in \mathfrak{d}(\mathfrak{A}) \cap \mathfrak{j}(\mathfrak{m}).$$

First we have $[D, L(u)] = [L(Du)] \in L(\mathfrak{m})$ for all $u \in \mathfrak{A}$. Using this result we find

$$[D, [L(u), L(v)]] \subset [L(\mathfrak{A}), [D, L(\mathfrak{A})]] \subset [L(\mathfrak{A}), L(\mathfrak{m})] \subset \mathfrak{i}(\mathfrak{m})$$

b) Part a) yields in particular $[\mathfrak{j}(\mathfrak{m}), \mathfrak{j}(\mathfrak{m})] \subset \mathfrak{i}(\mathfrak{m})$. Thus an induction on k proves the statement.

2. Next we define for any two ideal $\mathfrak{m}_1$ and $\mathfrak{m}_2$ in $\mathfrak{A}$ a subspace of $\mathfrak{A}$ by setting

$$\mathfrak{m}_1 * \mathfrak{m}_2 = \mathfrak{i}(\mathfrak{m}_1)\mathfrak{m}_2 + \mathfrak{i}(\mathfrak{m}_2)\mathfrak{m}_1.$$

Using (4.1) we obtain

$$\mathfrak{m}_1 * \mathfrak{m}_2 = \mathfrak{m}_1\mathfrak{m}_2 + (\mathfrak{m}_1\mathfrak{m}_2)\mathfrak{A}. \tag{4.3}$$

Consequently $\mathfrak{m}_1\mathfrak{m}_2 = 0$ is equivalent to $\mathfrak{m}_1 * \mathfrak{m}_2 = 0$.

As in § 2.3 we define the ideals $\tilde{\mathfrak{J}}(\mathfrak{m}) = \mathfrak{i}(\mathfrak{m}) \oplus \mathfrak{m} \oplus \overline{\mathfrak{m}}$ and $\mathfrak{J}(\mathfrak{m}) = \mathfrak{j}(\mathfrak{m}) \oplus \mathfrak{m} \oplus \overline{\mathfrak{m}}$ of $\mathfrak{L}(\mathfrak{A})$. Since $\mathfrak{L}(\mathfrak{A})$ is a Lie algebra, the product of two ideals is again an ideal in $\mathfrak{L}(\mathfrak{A})$. Calculating for two ideal $\mathfrak{m}_1$ and $\mathfrak{m}_2$ of $\mathfrak{A}$.

$$[\tilde{\mathfrak{J}}(\mathfrak{m}_1), \tilde{\mathfrak{J}}(\mathfrak{m}_2)] = ([\mathfrak{i}(\mathfrak{m}_1), \mathfrak{i}(\mathfrak{m}_2)] + \mathfrak{m}_1 \square \mathfrak{m}_2 + \mathfrak{m}_2 \square \mathfrak{m}_1) \oplus \mathfrak{m}_1 * \mathfrak{m}_2 \oplus \overline{\mathfrak{m}_1 * \mathfrak{m}_2}. \tag{4.4}$$

Hence in particular from § 1,2 follows *that* $\mathfrak{m}_1 * \mathfrak{m}_2$ *is an ideal in* $\mathfrak{A}$. In the literature a construction like this one occurs for $\mathfrak{m}_1 = \mathfrak{m}_2$ (see A. J. Penico [6] and N. Jacobson [3], Chapter IV). It seems that the connection given by (4.4) illustrates very well why $\mathfrak{m}_1 * \mathfrak{m}_2$ has to be considered. As a trivial consequence (4.4) we obtain

$$\tilde{\mathfrak{J}}(\mathfrak{m}_1 * \mathfrak{m}_2) \subset [\tilde{\mathfrak{J}}(\mathfrak{m}_1), \tilde{\mathfrak{J}}(\mathfrak{m}_2)] \subset \mathfrak{J}(\mathfrak{m}_1 * \mathfrak{m}_2). \tag{4.5}$$

The inclusion on the right hand side is not the sharpest one can get, but it is enough for our purpose. Next we consider the ideals $\tilde{\mathfrak{J}}(\mathfrak{m})$ and $\mathfrak{J}(\mathfrak{m})$ in

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LEMMA 9. *Let* $\mathfrak{m}$ *be an ideal in* $\mathfrak{A}$. *Then*

a) $[\mathfrak{F}(\mathfrak{m}), \mathfrak{L}(\mathfrak{A})] \subset \tilde{\mathfrak{F}}(\mathfrak{m})$,

b) $(\mathfrak{F}(\mathfrak{m}))^{\langle k+1\rangle} \subset (\tilde{\mathfrak{F}}(\mathfrak{m}))^{\langle k\rangle} \subset (\mathfrak{F}(\mathfrak{m}))^{\langle k\rangle}$.

COROLLARY. $\tilde{\mathfrak{F}}(\mathfrak{m})$ *is solvable if and only if* $\mathfrak{F}(\mathfrak{m})$ *is solvable.*

Proof. It is enough to show $[\mathfrak{F}(\mathfrak{m}), \mathfrak{L}(\mathfrak{A})] \subset \tilde{\mathfrak{F}}(\mathfrak{m})$, as the other statement follows from this by induction on k. We calculate

$$(4.6) \quad [\mathfrak{F}(\mathfrak{m}), \mathfrak{L}(\mathfrak{A})] = ([\mathfrak{j}(\mathfrak{m}), \mathfrak{h}(\mathfrak{A})] + \mathfrak{m} \square \mathfrak{A} + \mathfrak{A} \square \mathfrak{m}) \oplus \mathfrak{n} \oplus \bar{\mathfrak{n}},$$

where $\mathfrak{n} = \mathfrak{h}(\mathfrak{A})\mathfrak{m} + \mathfrak{j}(\mathfrak{m})\mathfrak{A} \subset \mathfrak{m}$. But from (4.1) and Lemma 8a follows that the first component of (4.6) is contained in $\mathfrak{i}(\mathfrak{m})$.

Finally we set $\mathfrak{m}^{(0)} = \mathfrak{m}$ and

$$\mathfrak{m}^{(k+1)} = \mathfrak{m}^{(k)} * \mathfrak{m}^{(k)}.$$

Using (4.5) and Lemma 9 an induction on k leads to

$$(4.7) \qquad (\tilde{\mathfrak{F}}(\mathfrak{m}))^{\langle 2k-1\rangle} \subset \mathfrak{F}(\mathfrak{m})^{(k)}, \quad (\tilde{\mathfrak{F}}(\mathfrak{m}))^{\langle 2k\rangle} \subset \mathfrak{F}(\mathfrak{m})^{(k)}$$

for $k = 1, 2, \cdots$. Indeed the case $k = 1$ follows from (4.5) and

$$(\tilde{\mathfrak{F}}(\mathfrak{m}))^{\langle 2\rangle} = [\tilde{\mathfrak{F}}^{\langle 1\rangle}(\mathfrak{m}), \tilde{\mathfrak{F}}^{\langle 1\rangle}(\mathfrak{m})] \subset [\mathfrak{F}(\mathfrak{m}^{(1)}), \mathfrak{F}(\mathfrak{m})^{(1)})] \subset \tilde{\mathfrak{F}}(\mathfrak{m}^{(1)}).$$

On the other hand from (4.5) follows trivially

$$(4.8) \qquad \tilde{\mathfrak{F}}(\mathfrak{m}^{(k)}) \subset (\mathfrak{F}(\mathfrak{m}))^{\langle k\rangle}.$$

Combining (4.7) and (4.8) and using the corollaries of Lemma 8 and 9 as well as the fact that $(\mathfrak{j}(\mathfrak{m}))^{\langle k\rangle}$ is contained in $(\mathfrak{F}(\mathfrak{m}))^{\langle k\rangle}$, we end up with

THEOREM 5. *Let* $\mathfrak{m}$ *be an ideal in the Jordan algebra* $\mathfrak{A}$ *with unit element.*

Then the following statements are equivalent:

a) $\mathfrak{m}$ *is *-solvable, i.e. there exists an integer* k *such that* $\mathfrak{m}^{(k)} = 0$.

b) $\tilde{\mathfrak{F}}(\mathfrak{m})$ *is solvable,*

c) $\mathfrak{F}(\mathfrak{m})$ *is solvable.*

In this case both, $\mathfrak{i}(\mathfrak{m})$ *and* $\mathfrak{j}(\mathfrak{m})$ *are solvable in* $\mathfrak{h}(\mathfrak{A})$.

If the Jordan algebra $\mathfrak{A}$ has finite dimension, we may apply a theorem of A. J. Penico [6] (see also N. Jacobson [3], Chapter IV) that states the existence of an integer k such that $\mathfrak{m}^{(k)} \subset \mathfrak{m}^2$. Repeating this process we find

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for every r an integer $k = k(r)$ such that $\mathfrak{m}^{(k)} \subset \mathfrak{m}^{\langle r \rangle}$. Hence $\mathfrak{m}$ being solvable implies $\mathfrak{m}$ being *-solvable. Since the converse is also true we obtain the

COROLLARY. *If $\mathfrak{A}$ has finite dimension, then $\mathfrak{m}$ is solvable if and only if $\mathfrak{J}(\mathfrak{m})$ is solvable.*

3. Now let $\mathfrak{A}$ be a finite dimensional Jordan algebra with unit element. Let $\operatorname{Rad}\mathfrak{A}$ be the radical of $\mathfrak{A}$, i.e. the maximal solvable ideal, and let $\operatorname{Rad}\mathfrak{L}(\mathfrak{A})$ be the radical of the Lie algebra $\mathfrak{L}(\mathfrak{A})$, i.e. again the maximal solvable ideal.

THEOREM 6. *Let $\mathfrak{A}$ be a finite dimensional Jordan algebra with unit element. Then* $\operatorname{Rad}\mathfrak{L}(\mathfrak{A}) = \mathfrak{J}(\operatorname{Rad}\mathfrak{A})$.

Proof. Let $\mathfrak{M} = \operatorname{Rad}\mathfrak{L}(\mathfrak{A})$. Then the above Corollary yields $\mathfrak{J}(\operatorname{Rad}\mathfrak{A}) \subset \mathfrak{M}$. The image $\bar{\mathfrak{M}}$ of the solvable ideal $\mathfrak{M}$ under the main involution is also solvable and therefore contained in $\mathfrak{M}$. Thus $\bar{\mathfrak{M}} = \mathfrak{M}$. Splitting the ideal $\mathfrak{M}$ according to Lemma 1 we observe that $\mathfrak{M} = \mathfrak{g} \oplus \mathfrak{m} \oplus \bar{\mathfrak{m}}$, where $\mathfrak{m}$ is an ideal in $\mathfrak{A}$. The definition of $\tilde{\mathfrak{J}}(\mathfrak{m})$ and $\mathfrak{J}(\mathfrak{m})$ lead to $\tilde{\mathfrak{J}}(\mathfrak{m}) \subset \mathfrak{M} \subset \mathfrak{J}(\mathfrak{m})$. Since $\mathfrak{M}$ is solvable, the ideal $\tilde{\mathfrak{J}}(\mathfrak{m})$ is solvable. Theorem 5 yields that $\mathfrak{J}(\mathfrak{m})$ is solvable. Thus the above Corollary shows that $\mathfrak{m}$ is solvable too. Hence $\mathfrak{m} \subset \operatorname{Rad}\mathfrak{A}$ and $\mathfrak{M} \subset \mathfrak{J}(\operatorname{Rad}\mathfrak{A})$.

COROLLARY 1. *$\mathfrak{L}(\mathfrak{A})$ is semi-simple if and only if $\mathfrak{A}$ is semi-simple.*

As an abbreviation we shall use the following notation. Let $\mathfrak{B}$ be an arbitrary algebra. Then we write $\mathfrak{B} = \mathfrak{B}_1 \dotplus \mathfrak{B}_0$ if and only if $\mathfrak{B}_1$ is a subalgebra of $\mathfrak{B}$, $\mathfrak{B}_0$ is an ideal of $\mathfrak{B}$ and the sum is a direct sum of the vector spaces.

Now let $\mathfrak{A}$ be a Jordan algebra of finite dimension over a field of characteristic 0. Then by the theorem of Levi one has a decomposition $\mathfrak{L}(\mathfrak{A}) = \mathfrak{S} \dotplus \operatorname{Rad}\mathfrak{L}(\mathfrak{A})$, where $\mathfrak{S}$ is a semi-simple subalgebra of $\mathfrak{L}(\mathfrak{A})$. Using $\mathfrak{L}(\mathfrak{A}) = [\mathfrak{L}(\mathfrak{A}), \mathfrak{L}(\mathfrak{A})]$ and Lemma 9a we obtain $\mathfrak{L}(\mathfrak{A}) \subset \mathfrak{S} \dotplus \tilde{\mathfrak{J}}(\operatorname{Rad}\mathfrak{A})$. Hence $\tilde{\mathfrak{J}}(\operatorname{Rad}\mathfrak{A})$ cannot be properly contained in $\mathfrak{J}(\operatorname{Rad}\mathfrak{A})$.

COROLLARY 2. *Let $\mathfrak{A}$ be a finite dimensional Jordan algebra over a field of characteristic 0. Then* $\tilde{\mathfrak{J}}(\operatorname{Rad}\mathfrak{A}) = \mathfrak{J}(\operatorname{Rad}\mathfrak{A}) = \operatorname{Rad}\mathfrak{L}(\mathfrak{A})$.

4. Now let $\mathfrak{a}$ be a subalgebra of the Jordan algebra $\mathfrak{A}$ and let $\mathfrak{m}$ be an ideal in $\mathfrak{A}$ such that $\mathfrak{A} = \mathfrak{a} \dotplus \mathfrak{m}$. Suppose that $\mathfrak{a}$ contains the unit element of $\mathfrak{A}$. We write the elements $u \in \mathfrak{A}$ in the form $u = a + m$, where $a \in \mathfrak{a}$ and $m \in \mathfrak{m}$. Denote by $\mathfrak{s}(\mathfrak{a})$ the subalgebra of $\mathfrak{h}(\mathfrak{A})$ generated by $L(a)$, $a \in \mathfrak{a}$. Thus from $L(u) = L(a) + L(m)$ follows that

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$$\mathfrak{h}(\mathfrak{A}) = \mathfrak{s}(\mathfrak{a}) + \mathfrak{i}(\mathfrak{m})$$

is satisfied. In order to prove

$$\mathfrak{s}(\mathfrak{a}) \cap \mathfrak{i}(\mathfrak{m}) \subset \mathfrak{j}(\mathfrak{m}^*\mathfrak{m}) \tag{4.9}$$

we start with an element $L(a) + D$, $a \in \mathfrak{a}$, $D \in [L(\mathfrak{a}), L(\mathfrak{a})]$ in $\mathfrak{s}(\mathfrak{a})$. This element belongs to $\mathfrak{i}(\mathfrak{m})$ if $a \in \mathfrak{m}$ and $D \in [L(\mathfrak{m}), L(\mathfrak{A})]$. Thus $a = 0$ and $Dx = 0$ for all $x \in \mathfrak{a}$. But $D \in [L(\mathfrak{m}), L(\mathfrak{A})]$ implies $D\mathfrak{A} = D\mathfrak{m} \subset \mathfrak{m}^*\mathfrak{m}$.

Using (4.9) we obtain $\mathfrak{h}(\mathfrak{A}) = \mathfrak{s}(\mathfrak{a}) \dotplus \mathfrak{i}(\mathfrak{m})$ whenever $\mathfrak{m}^2 = 0$. Putting $\mathfrak{S}(\mathfrak{a}) = \mathfrak{s}(\mathfrak{a}) \oplus \mathfrak{a} \oplus \bar{\mathfrak{a}}$ we obtain a subalgebra $\mathfrak{S}(\mathfrak{a})$ of $\mathfrak{L}(\mathfrak{A})$ and it follows $\mathfrak{L}(\mathfrak{A}) = \mathfrak{S}(\mathfrak{a}) + \tilde{\mathfrak{J}}(\mathfrak{m})$. But in general the sum is not direct. However we have

$$\mathfrak{L}(\mathfrak{A}) = \mathfrak{S}(\mathfrak{a}) \dotplus \tilde{\mathfrak{J}}(\mathfrak{m}), \text{ whenever } \mathfrak{A} = \mathfrak{a} \dotplus \mathfrak{m} \text{ and } \mathfrak{m}^2 = 0. \tag{4.10}$$

LEMMA 10. *Let $\mathfrak{A}$ be a finite dimensional Jordan algebra with unit element over a field of characteristic 0. Suppose $\mathfrak{A} = \mathfrak{a} \dotplus \mathfrak{m}$, where $\mathfrak{a}$ is semi-simple and $\mathfrak{m}^2 = 0$. Then $\mathfrak{S}(\mathfrak{a})$ is semi-simple and $\mathfrak{J}(\mathfrak{m}) = \tilde{\mathfrak{J}}(\mathfrak{m})$.*

Proof. From $\mathfrak{A}/\mathfrak{m} \cong \mathfrak{a}$ we observe $\operatorname{Rad}\mathfrak{A} \subset \mathfrak{m}$. But $\mathfrak{m}^2 = 0$ implies $\mathfrak{m} = \operatorname{Rad}\mathfrak{A}$. Next we apply Corollary 2 of Theorem 6 which leads to $\tilde{\mathfrak{J}}(\mathfrak{m}) = \mathfrak{J}(\mathfrak{m}) = \operatorname{Rad}\mathfrak{L}(\mathfrak{A})$. Hence (4.10) implies $\mathfrak{L}(\mathfrak{A}) = \mathfrak{S}(\mathfrak{a}) \dotplus \operatorname{Rad}\mathfrak{L}(\mathfrak{A})$ and consequently $\mathfrak{S}(\mathfrak{a})$ has to be semi-simple.

Conversely we consider a decomposition $\mathfrak{L}(\mathfrak{A}) = \mathfrak{S} \dotplus \mathfrak{M}$, where $e \in \mathfrak{S}$ and where $\mathfrak{S}$ and $\mathfrak{M}$ are mapped into themselves under the main involution. Thus we get $\bar{e} \in \mathfrak{S}$ and $2Id = e \,\square\, e = [e, \bar{e}] \in \mathfrak{S}$. Using the same arguments as in the proof of Lemma 1 and using $\mathfrak{S} = \bar{\mathfrak{S}}$ we see that the subalgebra $\mathfrak{S}$ splits, i. e. there exists subspaces $\mathfrak{h}$ of $\mathfrak{h}(\mathfrak{A})$ and $\mathfrak{a}$ of $\mathfrak{A}$ such that $\mathfrak{S} = \mathfrak{h} \oplus \mathfrak{a} \oplus \bar{\mathfrak{a}}$. Using Lemma 1 we have also $\mathfrak{M} = \mathfrak{g} \oplus \mathfrak{m} \oplus \bar{\mathfrak{m}}$ where $\mathfrak{g}$ is an ideal in $\mathfrak{h}(\mathfrak{A})$ and $\mathfrak{m}$ an ideal in $\mathfrak{A}$. From $\mathfrak{S} + \mathfrak{M} = \mathfrak{L}(\mathfrak{A})$ now follows $\mathfrak{a} + \mathfrak{m} = \mathfrak{A}$ and the sum is direct. In order to show that $\mathfrak{a}$ is a subalgebra of $\mathfrak{A}$ we start with $a, b \in \mathfrak{a}$. Thus $a, \bar{b} \in \mathfrak{S}$ and $2ab = (a \,\square\, b)e = [a \,\square\, b, e] = [[a, \bar{b}], e] \in \mathfrak{S}$. Hence we have

$$\mathfrak{A} = \mathfrak{a} \dotplus \mathfrak{m} \text{ whenever } \mathfrak{L}(\mathfrak{A}) = \mathfrak{S} \dotplus \mathfrak{M}, e \in \mathfrak{S}, \bar{\mathfrak{S}} = \mathfrak{S}, \bar{\mathfrak{M}} = \mathfrak{M}, \tag{4.11}$$

and where $\mathfrak{a} = A\mathfrak{S}$ and $\mathfrak{m} = A\mathfrak{M}$.

Substituting $\mathfrak{M} = \operatorname{Rad}\mathfrak{L}(\mathfrak{A})$ and using Theorem 6 we obtain

THEOREM 7. *Let $\mathfrak{A}$ be a finite dimensional Jordan algebra with unit element e. Then $\mathfrak{A}$ admits a Wedderburn-decomposition $\mathfrak{A} = \mathfrak{a} \dotplus \operatorname{Rad}\mathfrak{A}$ whenever $\mathfrak{L}(\mathfrak{A})$ admits a Levi-decomposition $\mathfrak{L}(\mathfrak{A}) = \mathfrak{S} \dotplus \operatorname{Rad}\mathfrak{L}(\mathfrak{A})$, where $e \in \mathfrak{S} = \bar{\mathfrak{S}}$.*

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The subalgebra $\mathfrak{T}$ of $\mathfrak{L}(\mathfrak{A})$ spanned by the elements e, $\bar{e}$ and Id is semi-simple and invariant under the main involution of $\mathfrak{L}(\mathfrak{A})$. Using a theorem of E. J. Taft [7] we end up with a new proof of the Wedderburn principal theorem for Jordan algebras:

COROLLARY. *Let $\mathfrak{A}$ be a Jordan algebra with unit element and of finite dimension over a field of characteristic 0. Then there exists a semi-simple subalgebra $\mathfrak{a}$ of $\mathfrak{A}$ such that $\mathfrak{A} = \mathfrak{a} \dotplus \operatorname{Rad} \mathfrak{A}$.*

5. Let $\mathfrak{a}$ be a Jordan algebra with unit element e and let $\mathfrak{m}$ be a unital Jordan $\mathfrak{a}$-module (see N. Jacobson [3], Chapter III), i.e. $\mathfrak{a}$ acts on the vector space $\mathfrak{m}$ and if one defines a product in the direct sum $\mathfrak{A} = \mathfrak{a} + \mathfrak{m}$ of the vector spaces by

$$(a_1 + m_1)(a_2 + m_2) = a_1a_2 + a_1m_2 + a_2m_1, \qquad a_i \in \mathfrak{a}, m_i \in \mathfrak{m},$$

then one gets a Jordan algebra $\mathfrak{A}$ with unit element e, the *split zero extension of $\mathfrak{a}$ by $\mathfrak{m}$*. The module $\mathfrak{m}$ is now an ideal in $\mathfrak{A}$ such that $\mathfrak{m}^2 = 0$. Furthermore we have $\mathfrak{A} = \mathfrak{a} \dotplus \mathfrak{m}$. Hence from (4.10) follows

$$\mathfrak{L}(\mathfrak{A}) = \mathfrak{S} \dotplus \mathfrak{M}, \text{ where } \mathfrak{S} = \mathfrak{S}(\mathfrak{a}), \mathfrak{M} = \tilde{\mathfrak{J}}(\mathfrak{m}). \tag{4.12}$$

But in general we cannot conclude $[\mathfrak{M}, \mathfrak{M}] = 0$. However, since M is an ideal in $\mathfrak{L}(\mathfrak{A})$, $\mathfrak{M}$ is a $\mathfrak{S}$-module.

THEOREM 8. *Let $\mathfrak{a}$ be a Jordan algebra and suppose that every finite dimensional Lie $\mathfrak{S}(\mathfrak{a})$-module is completely reducible. Then every finite dimensional Jordan $\mathfrak{a}$-module $\mathfrak{m}$ is completely reducible.*

Proof. Using (4.12) we know that $\mathfrak{M} = \tilde{\mathfrak{J}}(\mathfrak{m})$ is completely reducible. Let $\mathfrak{n}$ be an $\mathfrak{a}$-invariant subspace of $\mathfrak{m}$, then $\mathfrak{n}$ is an ideal in $\mathfrak{A} = \mathfrak{a} + \mathfrak{m}$. Hence $\tilde{\mathfrak{J}}(\mathfrak{n})$ is an ideal in $\mathfrak{L}(\mathfrak{A})$ that is contained in $\tilde{\mathfrak{J}}(\mathfrak{m})$. In particular $\tilde{\mathfrak{J}}(\mathfrak{n})$ is an $\mathfrak{S}(\mathfrak{a})$-invariant subspace of $\tilde{\mathfrak{J}}(\mathfrak{m})$. By assumption there exists an invariant subspace $\mathfrak{N}'$ of $\tilde{\mathfrak{J}}(\mathfrak{m})$ such that $\tilde{\mathfrak{J}}(\mathfrak{m}) = \tilde{\mathfrak{J}}(\mathfrak{n}) + \mathfrak{N}'$ and $\tilde{\mathfrak{J}}(\mathfrak{n}) \cap \mathfrak{N}' = 0$. Analogous to § 1.2 we obtain the fact that $\mathfrak{n}' = A\mathfrak{N}'$ is invariant under $\mathfrak{S}(\mathfrak{a})$ and consequently under $\mathfrak{a}$. Moreover it follows that $\mathfrak{m} = \mathfrak{n} + \mathfrak{n}'$ and this sum is direct, because of the fact that the ideal $\mathfrak{N}'$ of $\mathfrak{L}(\mathfrak{A})$ splits (see Lemma 1).

If $\mathfrak{a}$ is a finite dimensional semi-simple Jordan algebra then $\mathfrak{S}(\mathfrak{a})$ is a finite dimensional semi-simple (see Lemma 10) Lie algebra. Since over a field of characteristic 0 every finite dimensional Lie module with respect to a semi-simple Lie algebra is completely reducible we obtain the

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COROLLARY. *Let* $\mathfrak{a}$ *be a finite dimensional semi-simple Jordan algebra over a field of characteristic* 0. *Then every finite dimensional Jordan* $\mathfrak{a}$*-module is completely reducible.*

This is a special case of a theorem due to F. D. Jacobson and N. Jacobson [2].

5. Associative domains for Jordan algebras.

1. Let $\mathfrak{B}$ be an associative algebra with product $x \cdot y$ and commutator $[x, y] = x \cdot y - y \cdot x$. We consider the polynomial

$$k(x, y) = [[x, y], [x, [y, [x, y]]]]$$

for $x, y \in \mathfrak{B}$, that is symmetric in x and y because of the Jacobi identity. Let $\mathfrak{A}$ be a subspace of $\mathfrak{B}$ and $q \in \mathfrak{B}$ such that any two elements of $\mathfrak{A}$ commute in the multiplication of $\mathfrak{B}$ and $[a, [q, b]] \in \mathfrak{A}$ whenever $a, b \in \mathfrak{A}$. Setting

$$ab = [a, [q, b]] \tag{5.1}$$

we define a new product in the space $\mathfrak{A}$. Again the Jacobi identity yields $ab = ba$ and hence $\mathfrak{A}$ can be considered as a commutative algebra with respect to the product (5.1).

Following a proof of M. Osborn we show that in the product of $\mathfrak{A}$

$$a^2(ab) - a(a^2b) = [b, k(a, q)], \qquad a, b \in \mathfrak{A}, \tag{5.2}$$

is satisfied. First we have

$$\begin{aligned}(ab)c - a(bc) &= c[b, [q, a]] - a[b, [q, c]] \\ &= [[b, [q, a]], [q, c]] - [[b, [q, c]], [q, a]] \\ &= [b, [[q, a], [q, c]]].\end{aligned}$$

Consequently it follows $a^2(ab) - a(a^2b) = [b, u]$, where $u = [[q, a], [q, a^2]]$. But $[q, a^2] = [q, [a, [q, a]]] = [a, [q, [q, a]]]$ yields $u = k(a, q)$.

These considerations lead to the following definition: A pair $(\mathfrak{B}, q)$ of an associative algebra $\mathfrak{B}$ and $q \in \mathfrak{B}$ is called *an associative domain for the algebra* $\mathfrak{A}$, *if*

(*) $\mathfrak{A}$ is a subspace of $\mathfrak{B}$ and any two elements of $\mathfrak{A}$ commute in the multiplication of $\mathfrak{B}$,

(**) the product of $\mathfrak{A}$ is given by $ab = [a, [q, b]]$,

(***) $k(a, q) = 0$ for all $a \in \mathfrak{A}$.

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From (**) follows that $\mathfrak{A}$ is necessarily commutative and (***) together with (5.2) yields that $\mathfrak{A}$ is a Jordan algebra.

2. Conversely, let $\mathfrak{A}$ be a Jordan algebra with unit element e. For an associative algebra $\mathfrak{B}$ we denote by $\mathfrak{B}^-$ the Lie algebra defined by the commutator in the same vector space as $\mathfrak{B}$. *Suppose that the Lie algebra* $\mathfrak{L}(\mathfrak{A})$ *is a subalgebra of* $\mathfrak{B}^-$ *for some associative algebra* $\mathfrak{B}$. From $[a, a] = 0$ for $a \in \mathfrak{A}$ we see that (*) is satisfied for $\mathfrak{A}$. From (2.4) and (1.7) it follows that

$$P(a)b = U_a\bar{b} = \tfrac{1}{2}[a, [\bar{b}, a]] \text{ for } a, b \in \mathfrak{A}. \tag{5.3}$$

Substituting $b = e$ we obtain $a^2 = \frac{1}{2}[a, [q, a]]$, where $q = \bar{e}$. Since $[a, [q, b]]$ is symmetric in a and b we obtain (**) by linearization. Using

$$[a, q] = a \square e = 2L(a) \text{ and } [a, \bar{a}] = a \square \bar{a} = 2L(a^2)$$

one verifies that

$$k(a, q) = 4[L(a), [a, [\bar{e}, L(a)]]] = 4[L(a), [a, \bar{a}]] = 8[L(a), L(a^2)] = 0.$$

Hence (***) is satisfied, too, and we end up with an associative domain $(\mathfrak{B}, \bar{e})$ for the Jordan algebra $\mathfrak{A}$.

We make two applications of these results. First we consider the universal enveloping algebra $\mathfrak{U}$ of the Lie algebra $\mathfrak{L}(\mathfrak{A})$ having the product $x \cdot y$. We may assume that $\mathfrak{L}(\mathfrak{A})$ and consequently $\mathfrak{A}$ are subspaces of $\mathfrak{U}$. Hence $\mathfrak{L}(\mathfrak{A})$ is a subalgebra of $\mathfrak{U}^-$ and we get

THEOREM 9. *Let* $\mathfrak{A}$ *be a Jordan algebra with unit element* e *and let* $\mathfrak{U}$ *be the universal enveloping algebra of the Lie algebra* $\mathfrak{L}(\mathfrak{A})$. *Then* $(\mathfrak{U}, \bar{e})$ *is an associative domain for* $\mathfrak{A}$.

On the other hand consider a finite dimensional Jordan algebra $\mathfrak{A}$. It is well known that there exists a faithful representation of $\mathfrak{A}$ in a finite dimensional associative algebra. Hence there exists a finite dimensional associative algebra $\mathfrak{B}$ such that $\mathfrak{L}(\mathfrak{A})$ is a subalgebra of $\mathfrak{B}^-$.

THEOREM 10. *Let* $\mathfrak{A}$ *be a finite dimensional Jordan algebra with unit element. Then there exists a finite dimensional associative domain for* $\mathfrak{A}$.

3. Let $\mathfrak{A}$ be a special Jordan algebra with unit element e. Without loss of generality we may assume that $\mathfrak{A}$ is a subalgebra of $\mathfrak{C}^+$ where $\mathfrak{C}$ is an associative algebra and where the product in $\mathfrak{C}^+$ is given by $u \circ v = \frac{1}{2}(uv + vu)$. Denote by $\mathfrak{C}^-$ the Lie algebra which is defined in the same vector space as $\mathfrak{C}$ having the product $[u, v] = uv - vu$.

For $c \in \mathfrak{C}$ we define $D(c) \in \mathrm{Hom}(\mathfrak{C}, \mathfrak{C})$ by $D(c)u = cu - uc$. From

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$D([a,b]) = 4[L(a), L(b)]$ follows that $D: [\mathfrak{A}, \mathfrak{A}] \to \mathfrak{d}(\mathfrak{A})$ is a surjective homomorphism of the Lie algebras. Since $\mathfrak{A}$ is closed under the product $u \circ v$ we obtain $[\mathfrak{A}, [\mathfrak{A}, \mathfrak{A}]] \subset \mathfrak{A}$ and therefore that $[\mathfrak{A}, \mathfrak{A}]$ is closed under the product $[u, v]$, hence indeed $[\mathfrak{A}, \mathfrak{A}]$ is a Lie algebra, i.e. a subalgebra of $\mathfrak{C}^-$.

We suppose now that $D: [\mathfrak{A}, \mathfrak{A}] \to \mathfrak{d}(\mathfrak{A})$ *is injective*, i.e. that there is no non-zero element in $[\mathfrak{A}, \mathfrak{A}]$ which commutes with all of $\mathfrak{A}$ in the product of $\mathfrak{C}$. Then we define a map $\rho: \mathfrak{h}(\mathfrak{A}) \to \mathfrak{C}^-$ by setting

$$\rho(L(a)) = \tfrac{1}{2}a, \qquad \rho(\Sigma[L(b), L(c)]) = \tfrac{1}{4}\Sigma[b, c], \qquad a, b, c \in \mathfrak{A}.$$

Since ρD is the identity on $[\mathfrak{A}, \mathfrak{A}]$, we see that ρ is well defined. One verifies *that* ρ *is an injective homomorphism of the Lie algebras.* Finally

$$\rho(a \square b) = ab, \qquad a, b \in A, \tag{5.4}$$

and

$$\rho(T)u + u\rho(T^*) = Tu, \qquad T \in \mathfrak{h}(\mathfrak{A}), u \in \mathfrak{A}, \tag{5.5}$$

is satisfied.

Next we define an injective map $\omega: \mathfrak{L}(\mathfrak{A}) \to \mathfrak{B}$, where $\mathfrak{B}$ is the associative algebra of 2 by 2 matrices with entries in $\mathfrak{C}$ by setting

$$\omega(T \oplus a \oplus \bar{b}) = \begin{pmatrix} \rho(T) & a \\ b & -\rho(T^*) \end{pmatrix}.$$

Using (5.4) and (5.5) one verifies that ω is a homomorphism of the Lie algebra $\mathfrak{L}(\mathfrak{A})$ into $\mathfrak{B}^-$. Uusing the element $q = \begin{pmatrix} 0 & 0 \\ \frac{1}{2}e & 0 \end{pmatrix}$ of $\mathfrak{B}$ and the identification of $a \in \mathfrak{A}$ with $\begin{pmatrix} 0 & a \\ 0 & 0 \end{pmatrix}$ we see, *that the pair* $(\mathfrak{B}, q)$ *is an associative domain for* $\mathfrak{A}$. Since any two elements of $\mathfrak{A}$ have product 0 in $\mathfrak{B}$, we conclude from (**) that $a \circ b = \frac{1}{2}(aqb + bqa)$.

6. Associative bilinear forms.

1. A symmetric bilinear form τ (with values in an extension field of our ground field) of an arbitrary algebra $\mathfrak{B}$ is called *associative* if $\tau(ab, c) = \tau(a, bc)$ for $a, b, c \in \mathfrak{B}$.

Let $\mathfrak{A}$ be a Jordan algebra[1] with unit element e and let σ be an associative bilinear form of $\mathfrak{A}$. First one verifies

[1] For Section 1 and 2 we only need the identity (B), i.e. the fact that any transformation $[L(a), L(b)]$ is a derivation of $\mathfrak{A}$.

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(6.1) $\sigma(Ta, b) = \sigma(a, T^*b)$ for $T \in \mathfrak{h}(\mathfrak{A}) = L(\mathfrak{A}) + [L(\mathfrak{A}), L(\mathfrak{A})]$

and second one gets

(6.2) $\sigma(c, [L(a), L(b)]d) = \sigma(a, [L(c), L(d)]b)$ for $a, b, d \in \mathfrak{A}$.

Using the associator $(a, b, c) = (ab)c - a(bc)$ one also verifies

(6.3) $\sigma(x, (a, b, c)) = \sigma(a, (x, c, b)) = \sigma(b, (c, x, a)) = \sigma(c, (b, a, x))$.

Since the elements D of $\mathfrak{d}(\mathfrak{A}) = [L(\mathfrak{A}), L(\mathfrak{A})]$ are derivations of the algebra $\mathfrak{A}$ we also have

(6.4) $D(a, b, c) = (Da, b, c) + (a, Db, c) + (a, b, Dc)$ for $a, b, c \in \mathfrak{A}$, $D \in \mathfrak{d}(\mathfrak{A})$.

De define a bilinear form $\sigma_{\mathfrak{d}}$ on the algebra $\mathfrak{d}(\mathfrak{A}) = [L(\mathfrak{A}), L(\mathfrak{A})]$ by setting

$$\sigma_{\mathfrak{d}}(D_1, D_2) = \Sigma\sigma(D_1a_2, b_2) = \Sigma\sigma(D_2a_1, b_1) = \Sigma\Sigma\{\sigma(b_1a_2, a_1b_2) - \sigma(a_1a_2, b_1b_2)\}$$

whenever

$$D_i = \Sigma[L(a_i), L(b_i)] \in \mathfrak{d}(\mathfrak{A})$$

From (6.2) follows that in this definition the second and the third terms coincide. Moreover σ is well-defined and symmetric on $\mathfrak{d}(\mathfrak{A})$.

LEMMA 11. *Let σ be an associative bilinear form on $\mathfrak{A}$. Then $\sigma_{\mathfrak{d}}$ is an associative bilinear form on the algebra $\mathfrak{d}(\mathfrak{A}) = [L(\mathfrak{A}), L(\mathfrak{A})]$.*

Proof. In the above notation one gets for a third element $C \in \mathfrak{d}(\mathfrak{A})$ $-[D_1, C] = \Sigma[C, [L(a_1), L(b_1)]] = \Sigma([L(Ca_1), L(b_1)] + [L(a_1), L(Cb_1)])$ and similarly

$$[C, D_2] = \Sigma([L(Ca_2), L(b_2)] + [L(a_2), L(Cb_2)]),$$

because C is a derivation of $\mathfrak{A}$. Therefore

$$\begin{aligned}
&-\sigma_{\mathfrak{d}}([D_1, C], D_2) + \sigma_{\mathfrak{d}}(D_1, [C, D_2]) \\
&\quad = \Sigma\Sigma\{\sigma([L(Ca_1), L(b_1)]a_2, b_2) + \sigma([L(a_1), L(Cb_1)]a_2, b_2) \\
&\qquad + \sigma([L(Ca_2), L(b_2)]a_1, b_1) + \sigma([L(a_2), L(Cb_2)]a_1, b_1)\} \\
&\quad = \Sigma\Sigma\{\sigma((b_1, a_2, Ca_1) + (Cb_1, a_2, a_1), b_2) \\
&\qquad\qquad + \sigma((b_2, a_1 Ca_2) + (Cb_2, a_1, a_2), b_1)\} \\
&\quad = \Sigma\Sigma\{\sigma((b_1, a_2, Ca_1) + (Cb_1, a_2, a_1) + (b_1, Ca_2, a_1), b_2) \\
&\qquad\qquad + \sigma((b_1, a_2, a_1), Cb_2)\}
\end{aligned}$$

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using (6.3). But each term of this sum vanishes because of (6.1), (6.4) and $C^* = -C$.

2. Next we consider the Lie algebra $\mathfrak{h}(\mathfrak{A}) = L(\mathfrak{A}) + [L(\mathfrak{A}), L(\mathfrak{A})]$ and we define a symmetric bilinear form $\sigma_{\mathfrak{h}}$ on $\mathfrak{h}(\mathfrak{A})$ by setting

$$\sigma_{\mathfrak{h}}(L(a_1) + D_1, L(a_2) + D_2) = \sigma_{\mathfrak{d}}(D_1, D_2) + \sigma(a_1, a_2) \text{ for } a_i \in \mathfrak{A}, D_i \in \mathfrak{d}(\mathfrak{A}).$$

Obviously, the restriction of $\sigma_{\mathfrak{h}}$ on $\mathfrak{d}(\mathfrak{A})$ coincides with $\sigma_{\mathfrak{d}}$ and $L(\mathfrak{A})$ and $\mathfrak{d}(\mathfrak{A})$ are orthogonal with respect to $\sigma_{\mathfrak{h}}$.

LEMMA 12. *Let σ be an associative bilinear form on $\mathfrak{A}$. Then $\sigma_{\mathfrak{h}}$ is an associative bilinear form of the Lie algebra $\mathfrak{h}(\mathfrak{A})$ satisfying $\sigma_{\mathfrak{h}}(a \square b, T) = 2\sigma(Ta, b)$ for $a, b \in \mathfrak{A}$ and $T \in \mathfrak{h}(\mathfrak{A})$.*

Proof. One verifies that $\sigma_{\mathfrak{h}}$ satisfied the above identity. Let

$$T_i = L(a_i) + D_i, \quad D_i \in \mathfrak{d}(\mathfrak{A}),$$

be elements of $\mathfrak{h}(\mathfrak{A})$. We calculate

$$\begin{aligned}\sigma_{\mathfrak{h}}([T_i, T_j], T_k) &= \sigma_{\mathfrak{h}}([L(a_i), L(a_j)] + L(D_i a_j - D_j a_i) + [D_i, D_j], L(a_k) + D_k) \\ &= \sigma_{\mathfrak{d}}([D_i, D_j], D_k) + \sigma(D_k a_i, a_j) + \sigma(D_i a_j, a_k) + \sigma(D_j a_k, a_i)\end{aligned}$$

using (6.1) and $D_j^* = -D_j$. Hence the right hand side does not change when i, j, k are permuted cyclically, and therefore $\sigma_{\mathfrak{h}}$ is associative.

3. Finally we define a symmetric bilinear form $\sigma_{\mathfrak{L}}$ of $\mathfrak{L}(\mathfrak{A})$ setting

$$\sigma_{\mathfrak{L}}(x_1, x_2) = \sigma_{\mathfrak{h}}(T_1, T_2) + 2\sigma(a_1, b_2) + 2\sigma(a_2, b_1)$$

whenever $x_i = T_i \oplus a_i \oplus \bar{b}_i \in \mathfrak{L}(\mathfrak{A})$. Using (6.1) and Lemma 12 one verifies that

$$\begin{aligned}\sigma_{\mathfrak{L}}([x_i, x_j], x_k) = \sigma_{\mathfrak{h}}([T_i, T_j], T_k) &+ 2\sigma(T_k a_i, b_j) - 2\sigma(T_k a_j, b_i) \\ &+ 2\sigma(T_i a_j, b_k) - 2\sigma(T_j a_i, b_k) + 2\sigma(T_j a_k, b_i) - 2\sigma(T_i a_k, b_j).\end{aligned}$$

Again, since the right hand side does not change when i, j, k are permuted cyclically, we obtain

THEOREM 11. *Let $\mathfrak{A}$ be a Jordan algebra with unit element and let σ be an associative bilinear form on $\mathfrak{A}$. Then $\sigma_{\mathfrak{L}}$ is an associative bilinear form on $\mathfrak{L}(\mathfrak{A})$. Moreover $\sigma \neq 0$ implies $\sigma_{\mathfrak{L}} \neq 0$.*

4. Let τ be a symmetric bilinear form of an arbitrary algebra $\mathfrak{B}$. Then the subspace

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$$Bk_\tau(\mathfrak{B}) = \{a; a \in \mathfrak{B}, \tau(a, b) = 0 \text{ for all } b \in \mathfrak{B}\}$$

is called the *bilinear kernel* of τ (or the radical of τ). If τ is associative then $Bk_\tau(\mathfrak{B})$ is an ideal of $\mathfrak{B}$.

Again let σ be an associative bilinear form of the Jordan algebra $\mathfrak{A}$. We shall prove that

$$Bk_{\sigma\mathfrak{h}}(\mathfrak{h}(\mathfrak{A})) = \mathfrak{j}(Bk_\sigma(\mathfrak{A})), \tag{6.5}$$

where $\mathfrak{j}(\mathfrak{m}) = \{T, T \in \mathfrak{h}(\mathfrak{A}), T\mathfrak{A} \subset \mathfrak{m}\}$. Put $\mathfrak{m} = Bk_\sigma(\mathfrak{A})$ and denote the left hand side of (6.5) by $\mathfrak{g}$. From the definition of $\sigma_\mathfrak{h}$ it follows that $L(a) + D$ belongs to $\mathfrak{g}$ if and only if $a \in \mathfrak{m}$ and $\sigma_\mathfrak{d}(D, T) = 0$ for all $T \in \mathfrak{d}(\mathfrak{A})$. Using the definition of $\sigma_\mathfrak{d}$ for $T = [L(c), L(d)]$ one can replace the last condition by $\sigma(Dc, d) = 0$ for all $c, d \in \mathfrak{A}$. Hence $L(a) + D \in \mathfrak{g}$ is equivalent to $a \in \mathfrak{m}$ and $D\mathfrak{A} \subset \mathfrak{m}$, and therefore, because of (4.2) to $L(a) + D \in \mathfrak{j}(\mathfrak{m})$. This proves (6.5).

Now from the definition of $\sigma_\mathfrak{L}$ it follows that the bilinear kernel of $\mathfrak{L}(\mathfrak{A})$ with respect to $\sigma_\mathfrak{L}$ is equal to $\mathfrak{g} \oplus \mathfrak{m} \oplus \bar{\mathfrak{m}}$ and we end up with

$$Bk_{\sigma\mathfrak{L}}(\mathfrak{L}(\mathfrak{A})) = \mathfrak{J}(Bk_\sigma(\mathfrak{A})). \tag{6.6}$$

As a first consequence we see *that $\sigma_\mathfrak{L}$ is non degenerate if σ is.*

For another application *we suppose that $\mathfrak{A}$ is finite dimensional.* From [1], Chapter III, Satz 5.2b, and Chapter I, Lemma 6.1, follows that there are non-zero associative bilinear forms on $\mathfrak{A}$. *Hence there exists non-zero associative bilinear forms of the Lie algebra $\mathfrak{L}(\mathfrak{A})$.* Since the radical $\operatorname{Rad}\mathfrak{A}$ of $\mathfrak{A}$ coincides with the intersction of the bilinear kernels of all seminormal bilinear forms of $\mathfrak{A}$ (see [1], Chapter 1, § 7, and III, Satz 8.4) we conclude from (6.6) and Theorem 6 *that the radical of the Lie algebra $\mathfrak{L}(\mathfrak{A})$ is the intersection of the bilinear kernels of certain associative bilinear forms on $\mathfrak{L}(\mathfrak{A})$*, namely those bilinear form $\sigma_\mathfrak{L}$ for which σ is seminormal on $\mathfrak{A}$. It would be of interest to characterize these bilinear forms in terms of $\mathfrak{L}(\mathfrak{A})$.

5. We still suppose that $\mathfrak{A}$ is finite dimensional and we denote by $\langle\ ,\ \rangle_\mathfrak{S}$ the *Killing form* of the Lie algebra $\mathfrak{S}$, whenever $\mathfrak{S}$ is a subalgebra of $\mathfrak{L}(\mathfrak{A})$. For $y \in \mathfrak{h}(\mathfrak{A})$ resp. $y \in \mathfrak{A} \oplus \bar{\mathfrak{A}}$ the map $x \to [y, [y, x]]$ of $\mathfrak{A} \oplus \bar{\mathfrak{A}}$ into itself is denoted by $M(y)$. Using [1], Chapter IX, Lemma 5.2, we obtain

$$\langle x, x\rangle_{\mathfrak{L}(\mathfrak{A})} = \langle T, T\rangle_{\mathfrak{h}(\mathfrak{A})} + \operatorname{Trace} M(T) + \cdots \operatorname{Trace} M(a \oplus \bar{b}),$$

whenever $x = T \oplus a \oplus \bar{b} \in \mathfrak{L}(\mathfrak{A})$. From

$$M(T)(u \oplus \bar{v}) = [T, [T, u \oplus \bar{v}]] = [T, Tu \oplus (-\overline{T^*v})] = T^2u \oplus \overline{T^{*2}v}$$

follows $\operatorname{Trace} M(T) = \operatorname{Trace} T^2 + \operatorname{Trace}(T^*)^2$. Similar from

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$$M(a \oplus \bar{b})(u \oplus \bar{v}) = [a \oplus \bar{b}, [a \oplus \bar{b}, u \oplus \bar{v}]]$$
$$= [a \oplus \bar{b}, a \square v - u \square b] = (u \square b - a \square v)a \oplus \overline{(v \square a - b \square u)b}$$
$$= [(a \square b)u - P(a)v] \oplus \overline{[(b \square a)v - P(b)u]}$$

follows $\operatorname{Trace} M(a \oplus \bar{b}) = \operatorname{Trace}\ a \square b + \operatorname{Trace} b \square a = 4 \operatorname{Trace} L(ab)$.

Summing up we obtain for the Killing form of $\mathfrak{L}(\mathfrak{A})$

$$(6.7) \quad \langle x_1, x_2 \rangle_{\mathfrak{L}(\mathfrak{A})} = \langle T_1, T_2 \rangle_{\mathfrak{h}(\mathfrak{A})} + \operatorname{Trace} T_1 T_2 + \operatorname{Trace} T_1^* T_2^* + 4 \operatorname{Trace} L(a_1 b_2 + a_2 b_1)$$

whenever $x_i = T_i \oplus a_i \oplus \bar{b}_i$.

Setting

$$\tau(a, b) = \operatorname{Trace} L(ab), \quad a, b \in \mathfrak{A},$$

we get an associative bilinear form of $\mathfrak{A}$. Using τ we construct $\tau_{\mathfrak{h}}$ and $\tau_{\mathfrak{L}}$ as described in Section 1-3. According to (6.7) the bilinear form $\lambda(x_1, x_2) = 2\tau_{\mathfrak{L}}(x_1, x_2) - \langle x_1, x_2 \rangle_{\mathfrak{L}(\mathfrak{A})}$ vanishes on the subspace $\mathfrak{A} \oplus \bar{\mathfrak{A}}$ of $\mathfrak{L}(\mathfrak{A})$. Moreover, $\mathfrak{h}(\mathfrak{A})$ and $\mathfrak{A} \oplus \bar{\mathfrak{A}}$ are orthogonal with respect to λ, because they are orthogonal with respect to $\tau_{\mathfrak{L}}$ and to the Killing form. For $x_i = T_i \oplus a_i \oplus \bar{b}_i$ therefore it follows that $\lambda(x_1, x_2) = \lambda(T_1, T_2)$. Since $\mathfrak{L}(\mathfrak{A})$ is a Lie algebra, its Killing form is associative and therefore λ is associative on $\mathfrak{L}(\mathfrak{A})$. Hence we obtain $\lambda(u \square v, T_2) = \lambda([u, \bar{v}], T_2) = \lambda(u, [\bar{v}, T_2]) = 0$ because $[\bar{v}, T_2] \in \bar{\mathfrak{A}}$. But any element T_1 of $\mathfrak{h}(\mathfrak{A})$ is spanned by elements of the form $u \square v$, so we end up with $\lambda = 0$ or

$$(6.8) \quad \langle x_1, x_2 \rangle_{\mathfrak{L}(\mathfrak{A})} = 2\tau_{\mathfrak{L}}(x_1, x_2).$$

Again using (6.7) and the definition of $\tau_{\mathfrak{L}}$, we obtain

$$(6.9) \quad \langle T_1, T_2 \rangle_{\mathfrak{h}(\mathfrak{A})} = 2\tau_{\mathfrak{h}}(T_1, T_2) - \operatorname{Trace} T_1 T_2 - \operatorname{Trace} T_1^* T_2^*$$

and more explicitly for $T_i = L(a_i) + D_i$,

$$\tfrac{1}{2}\langle T_1, T_2 \rangle_{\mathfrak{h}(\mathfrak{A})} = \tau_{\mathfrak{d}}(D_1, D_2) - \operatorname{Trace} D_1 D_2 + \operatorname{Trace}\{L(a_1 a_2) - L(a_1) L(a_2)\}.$$

On the other hand analogous to [1], Chapter IX, Satz 5.3, it follows that

$$\tfrac{1}{2}\langle T_1, T_2 \rangle_{\mathfrak{h}(\mathfrak{A})} = \tfrac{1}{2}\langle D_1, D_2 \rangle_{\mathfrak{d}(\mathfrak{A})} + \tfrac{1}{2} \operatorname{Trace} D_1 D_2 + \operatorname{Trace}\{L(a_1 a_2) - L(a_1) L(a_2)\}.$$

Combining the last both equations we end up with

$$(6.10) \quad \langle D_1, D_2 \rangle_{\mathfrak{d}(\mathfrak{A})} = 2\tau_{\mathfrak{d}}(D_1, D_2) - 3 \operatorname{Trace} D_1 D_2.$$

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The formulas (6.8), (6.9) and (6.10) are explicit expressions for the Killing form because $\tau_{\mathfrak{L}}$, $\tau_{\mathfrak{h}}$ and $\tau_{\mathfrak{d}}$ are explicitly given.

Suppose now that $\mathfrak{A}$ is central simple and that the ground field has characteristic 0. Using a result of K. Meyberg [5] (see also [1], Chapter IX, Satz 3.6) we have

$$\langle D_1, D_2\rangle_{\mathfrak{d}(\mathfrak{A})} = \alpha(\mathfrak{A}) \operatorname{Trace} D_1 D_2, \tag{6.11}$$

where $\alpha(\mathfrak{A})$ is a non negative rational number. Hence (6.10) yields

$$\tau_{\mathfrak{d}}(D_1, D_2) = \tfrac{1}{2}(\alpha(\mathfrak{A}) + 3) \operatorname{Trace} D_1 D_2. \tag{6.11'}$$

Furthermore one gets

$$\begin{aligned} &\langle T_1, T_2\rangle_{\mathfrak{h}(\mathfrak{A})} \\ &= (\alpha(\mathfrak{A}) + 1) \operatorname{Trace} D_1 D_2 + 2 \operatorname{Trace}\{L(a_1 a_2) - L(a_1) L(a_2)\} \end{aligned} \tag{6.12}$$

$$\tau_{\mathfrak{h}}(T_1, T_2) = \tfrac{1}{2}(\alpha(\mathfrak{A}) + 3) \operatorname{Trace} D_1 D_2 + \operatorname{Trace} L(a_1 a_2). \tag{6.12'}$$

6. We use now the results of § 2.6. Let σ be an associative bilinear form of the Jordan algebra $\mathfrak{A}$ with unit element and let J be an automorphism of $\mathfrak{A}$ satisfying $J^2 = Id$ and

$$\sigma(Ja, b) = \sigma(a, Jb) \text{ for } a, b \in \mathfrak{A}. \tag{6.13}$$

We construct the associated bilinear form $\sigma_{\mathfrak{h}}$ of $\mathfrak{h}(\mathfrak{A})$ and shall prove that

$$\sigma_{\mathfrak{h}}(S^J, T) = \sigma_{\mathfrak{h}}(S, T^J) \text{ for } S, T \in \mathfrak{h}(\mathfrak{A}). \tag{6.14}$$

Since $L(\mathfrak{A})$ and $\mathfrak{d}(\mathfrak{A}) = [L(\mathfrak{A}), L(\mathfrak{A})]$ are orthogonal with respect to $\sigma_{\mathfrak{h}}$ and since $\sigma_{\mathfrak{h}}(L(a), L(b)) = \sigma(a, b)$ is satisfied, it is enough to prove (6.14) for $S, T \in \mathfrak{d}(\mathfrak{A})$. But in this case the statement follows from the definition of $\sigma_{\mathfrak{h}}$ in Section 1 and from (6.13).

Obviously, (6.14) yields *that* $\mathfrak{h}^+(\mathfrak{A}; J)$ *and* $\mathfrak{h}^-(\mathfrak{A}; J)$ *are orthogonal with respect to* $\sigma_{\mathfrak{h}}$, *whenever* (6.13) *is satisfied.*

Let $x_i = T_i \oplus a_i \oplus \epsilon_i \overline{Ja_i}$ be an element of $\mathfrak{L}^{\epsilon_i}$. Using (6.13) one verifies that

$$\sigma_{\mathfrak{L}}(x_1, x_2) = \sigma_{\mathfrak{h}}(T_1, T_2) + 2(\epsilon_1 + \epsilon_2)\sigma(a_1, Ja_2). \tag{6.15}$$

As a consequence we see *that* $\mathfrak{L}^+(\mathfrak{A}; J)$ *and* $\mathfrak{L}^-(\mathfrak{A}; J)$ *are orthogonal* with respect to $\sigma_{\mathfrak{L}}$, *whenever* (6.13) *is satisfied.*

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REFERENCES.

[1] H. Braun and M. Koecher, *Jordan-Algebren*, Springer Verlag, Berlin, 1966.

[2] F. D. Jacobson and N. Jacobson, "Classification and representation of semi-simple Jordan algebras," *Transactions of the American Mathematical Society*, vol. 65 (1949), pp. 141-169.

[3] N. Jacobson, *Jordan algebras* (a forthcoming book).

[4] J. L. Kantor, "Classification of irreducible transitively differential geometries," *Doklady Academi Nauk*, vol. 158 (1964), pp. 1271-1272.

[5] K. Meyberg, "Ueber die Killing-Form in Jordan-Algebren," *Mathematische Zeitschrift*, vol. 89 (1965), pp. 52-73.

[6] A. J. Penico, "The Wedderburn principal theorem for Jordan algebras," *Transactions of the American Mathematical Society*, vol. 70 (1951), pp. 404-420.

[7] E. J. Taft, "Invariant Levi factors," *Michigan Mathematical Journal*, vol. 9 (1962), pp. 65-68.

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